

Highlights

A Numerical Investigation of Indirect Adaptive Predictive Control with Structure-Informed Nonlinear Regressors

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- Supplied nonlinear structure is compared within a common PCAC architecture.
- Prediction, tracking, and parameter recovery are assessed separately.
- Physical identification is distinguished from nonlinear forward reconstruction.
- Affine freezing can reverse a nonlinear model's prediction advantage.

A Numerical Investigation of Indirect Adaptive Predictive Control with Structure-Informed Nonlinear Regressors

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Abstract

This paper numerically investigates nonlinear model construction from supplied structural information within predictive cost adaptive control. The study varies the retained nonlinear functions, coefficient sharing, and forward reconstruction under a common identification and control architecture. Recursive least squares updates unknown coefficients from direct or integrated relations that are linear in the parameters. The reconstructed nonlinear forward predictor need not be parameter-linear. A local affine approximation, constructed from the predictor value and Jacobian, is frozen over the prediction horizon for constrained quadratic optimization. Scalar full-state studies compare exact parameterizations, Taylor-informed approximations, broader dictionaries, and incorrectly restricted models. The comparisons include parameter variation, structural changes, and selected noisy cases. Common-data tests assess nonlinear prediction, whereas common-query forecasts distinguish nonlinear-model error from affine-freezing error. Under the tested conditions, correct structure improves prediction relative to affine models without increasing parameter count. Integrated physical identification reduces prediction error relative to direct sampled-map fitting after initialization. Additional coefficients and improved nonlinear prediction do not ensure lower tracking error. Forgetting improves readaptation without ensuring parameter recovery, and coefficient activation does not establish structure discovery. Affine freezing can reverse the nonlinear prediction

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advantage over an affine baseline. The results support evaluating representation, identification, reconstruction, and horizon prediction separately before increasing model complexity.

Keywords: Predictive cost adaptive control, Recursive least squares, Nonlinear system identification, Structure-informed models, Numerical verification

1. Introduction

Indirect adaptive predictive control identifies a model online and uses it for prediction and finite-horizon optimization. Predictive cost adaptive control (PCAC) combines recursive least-squares (RLS) identification with constrained receding-horizon optimization [1]. The PCAC predictive formulation is related to generalized predictive control [2, 3]. PCAC has been investigated for flutter suppression [4] and active flow control [5]. Experimental implementations include quadcopter flight [6] and acoustic noise suppression [7]. PCAC updates the model during operation to accommodate uncertainty and changes in the plant. For nonlinear plants, the design question concerns both the estimator and the structure supplied to the identification model. A complete numerical model may be unavailable even when some functional relations are known. This paper investigates how retaining such information affects prediction and closed-loop control.

Nonlinear identification includes models combining linear dynamics with static nonlinear functions [8], and noise-aware identification has also been studied for block-oriented nonlinear models [9, 10]. Finite nonlinear regressors permit models that are nonlinear in measured variables but linear in unknown coefficients [11]. The direct-history model in Sections 3.1 and 3.3 is a parameter-linear nonlinear autoregressive model with exogenous input (NARX), rather than a distinct identification class, and need not impose a Hammerstein cascade. Alhazmi et al. [12] identify pseudo-linear input-output models with finite polynomial, Fourier, and cubic-spline observables using RLS. Their iterative model predictive control (MPC) step updates the horizon prediction model using predicted trajectories. The present study compares model construction under a common Jacobian-frozen affine control step, without inner updates of the horizon model. The dictionary need not discard available plant information. Known contributions can remain fixed, and known nonlinear functions can be retained explicitly. Sharing unknowns

can impose coefficient relations without estimating a separate coefficient for every term. When the supplied functions and relations give an exact parameterization, identification concerns only the independent unknown parameters. Exact representability establishes that a suitable parameter vector exists, but does not establish that the available data identify that vector accurately.

For RLS, linearity in the unknown parameters is required of the identification relation, not of the reconstructed nonlinear forward predictor. The identification relation and forward predictor need not have the same form. Physical equations may supply an inverse or interval-integrated relation from which the parameters are estimated. Physical-identification methods exploit relations that are nonlinear in measured motion but linear in inertia entries [13]. Interval integration can avoid estimating state derivatives [14]. The scalar physical example uses an integrated parameter-linear relation and reconstructs the nonlinear forward map by propagating the identified differential equation. An exact physical parameterization does not make the quadrature or numerical forward propagation exact. Taylor expansion is unnecessary when a computable parameter-linear relation and a usable nonlinear forward model are available.

Partial structural information can restrict an approximate predictor when the exact functional form is unavailable. Taylor/Maclaurin structure provides one local construction under sufficient smoothness [15]. Known sparsity, symmetries, and couplings restrict the retained monomials and coefficient relations. The fitted coefficients need not equal the derivatives of the true map. Exact structural eliminations, finite-order truncation, and additional deliberate reduction are distinct. With little reliable functional information, a broader candidate dictionary remains an alternative, although its variables, memory, and functions must still be specified. A generic dictionary does not impose the particular structural restrictions being investigated. The amount of supplied information and the correctness of its restrictions are separate considerations. An approximation can retain valid structure, whereas an incorrect restriction can exclude dynamics needed for prediction.

Related linear parameter-varying (LPV) identification methods address unknown scheduling dependence. Tóth et al. [16] use least-squares support vector machines (LS-SVM) to estimate scheduling-dependent coefficient functions in LPV-ARX models through kernel representations. Rizvi et al. [17] first reconstruct a state sequence from input, output, and scheduling data using kernel canonical correlation analysis, and then identify scheduling-dependent state-space matrices with LS-SVM. Recursive LPV-ARX iden-

tification with informative-data selection is also available [18]. Here, RLS updates a fixed finite vector of independent coefficients in explicitly supplied regressors, including specified coefficient-sharing relations. Kernel choices can also incorporate prior information [16], and finite-feature kernel models can overlap with explicit regressions. The distinction is therefore the chosen parameterization and estimation procedure, not an exclusive ability to use prior information or update online. The functions and coefficient-sharing relations here are not selected online.

Other formulations of adaptive MPC combine recursive model updates with robust constraint treatment [19], RLS estimation with tube-based prediction [20], or uncertainty compensation with tightened constraints [21]. Buerger and Cannon combine successive linearization with ellipsoidal tubes in a robust adaptive nonlinear MPC formulation [22]. Explicit excitation constraints have also been used to support parameter identification [23]. The present study uses the same PCAC predictive-control step for all model constructions and does not propagate uncertainty sets or enforce excitation constraints. The controller evaluates and differentiates the reconstructed nonlinear map at the current operating point, using its value and Jacobian to define a local affine predictor. The affine offset is retained, so the operating point need not be an equilibrium. The affine coefficients are held fixed over the prediction horizon and recomputed at the next control instant. A quadratic tracking objective, hard input and input-increment bounds, and optional soft output bounds yield a constrained quadratic program (QP). Unlike nonlinear model predictive control [24], this QP does not optimize a recursive nonlinear rollout. Agreement between the nonlinear and affine maps at the current point does not establish agreement over the horizon. The approximation introduced by affine freezing is therefore assessed separately from nonlinear-model error. Predicted output bounds do not certify bounds on the actual plant output.

This paper contributes a structured numerical investigation of nonlinear regressor construction under the common PCAC architecture. It does not propose a new nonlinear-regression class or local-linearization principle. The comparisons assess how supplied structure affects nonlinear prediction and closed-loop tracking, while evaluating parameter recovery separately. Reference [15] develops Taylor-informed polynomial models with Jacobian-frozen affine predictors and compares polynomial degrees. Here, the Taylor-informed model is one choice alongside exact-function and physical-reconstruction models. We vary which functions are retained, which coeffi-

cients are estimated independently or shared, and how the nonlinear forward predictor is reconstructed from the estimates. The comparisons distinguish exact parameterizations, structure-informed approximations, broader candidate dictionaries, and incorrectly restricted models, with affine PCAC and known-model references. Stationary studies separate the value of supplied structure from the number of estimated coefficients and examine approximation order over prescribed operating regions. Further studies consider parameter drift, abrupt changes, and nonlinear plant changes within and outside the retained model class. A physical example compares integrated identification and nonlinear reconstruction with direct sampled-map fitting.

Common-data identification tests use common fitting records and independent evaluation records. Closed-loop tests assess adaptive controllers whose actions may generate different identification data. Forecast diagnostics use common initial conditions and input sequences to separate nonlinear-model error, affine-freezing error, and total prediction error over several horizons. Tracking, control activity, and readaptation are evaluated separately from parameter recovery. Selected comparisons include measurement noise and process disturbances, and numerical conditioning is monitored. RLS is compared without forgetting, with fixed forgetting, and with residual-triggered variable-rate forgetting [25]. The dictionaries remain fixed. Forgetting reweights past data but does not generate excitation or supply missing functions. Changes in fitted nonlinear coefficients are therefore not treated as evidence of online structure discovery.

The numerical studies use scalar sampled recurrences and a scalar differential equation with full-state measurements. This design isolates model construction and adaptation from state-estimation errors. The formulation also admits finite measured histories, but a chosen history length does not establish an exact representation of an arbitrary partially measured plant. State estimation and sensor realization are outside the scope. The known-model references retain the same frozen-affine QP and are diagnostic comparisons, not performance bounds. The investigation does not establish general closed-loop stability or parameter convergence. Section 5.3 gives a feasible-point construction for the implemented QP under stated conditions. The aim is to provide conditional design guidance by distinguishing limitations of representation, identification, reconstruction, and horizon prediction before introducing additional model complexity.

2. Control Problem and Available Information

2.1. Plant and available information

Consider a causal nonlinear plant with input $u_k \in \mathbb{R}^m$ and measured output $y_k \in \mathbb{R}^p$. Let $T_s > 0$ be the sampling period, and let $t_k = kT_s$ denote the sampling instants. In the absence of disturbances and measurement noise, denote the plant's input-output relation by

$$y_{k+1} = \mathcal{P}_k(y_{-\infty:k}, u_{-\infty:k}), \quad (1)$$

where $y_{-\infty:k}$ and $u_{-\infty:k}$ are the output and input histories through index k , respectively. The unknown operator $\mathcal{P}_k$ may depend on k , which permits but does not require time-varying plant dynamics. Equation (1) does not prescribe a finite memory or a state-space realization.

A complete numerical plant model is not supplied, although structural information about the plant may be supplied. A plant equilibrium is not assumed known.

2.2. Prediction model and control objective

The controller uses a finite-dimensional state-space model to predict future outputs. Let $\xi \in \mathbb{R}^n$ denote the state variable of this model. At each control instant t_k , the initial state ξ_k of the prediction trajectory is constructed from data available through t_k . The two constructions considered here use either a finite measured input-output history or measurements of the full physical state.

For the history construction, choose an integer $\ell \geq 1$ and define

$$\xi_k = s_k := \text{col}(y_k, \dots, y_{k-\ell+1}, u_{k-1}, \dots, u_{k-\ell+1}), \quad (2)$$

where $n = p\ell + m(\ell - 1)$. The vector s_k stores measured outputs and past inputs, with an empty input block when $\ell = 1$. The required history is supplied or collected before predictive control begins. This construction does not require a physical state-space model. Selecting ℓ specifies the memory of the prediction model but does not establish that the plant admits an exact representation using this history.

For the full-state construction only, assume that the measured output contains all components of a physical plant state. Set $n = p$, let $x_k \in \mathbb{R}^n$ denote that state, and write

$$y_k = x_k + v_k,$$

where $v_k \in \mathbb{R}^n$ is measurement noise. The prediction trajectory starts from the measured state,

$$\xi_k = y_k,$$

so $\xi_k = x_k$ in the noise-free case. No state observer is introduced.

For either construction, define the fixed output-selection matrix $C \in \mathbb{R}^{p \times n}$ to select the output coordinates from the prediction state. For the history construction, $C = [I_p \ 0]$, whereas for the full-state construction, $C = I_n$. At initialization, the measured output satisfies

$$y_k = C\xi_k. \tag{3}$$

The horizon starts from $\xi_{0|k} = \xi_k$, and the predicted outputs are $y_{i|k} = C\xi_{i|k}$.

During prediction, the history model predicts a new output and shifts the stored history through known shift equations. The full-state model propagates the modeled physical-state dynamics. The nonlinear state recursion for each construction is defined in the following section.

Let $r_k \in \mathbb{R}^p$ be the output reference. The control objective is to track r_k while respecting prescribed input and input-increment bounds.

At each control instant t_k , the input u_k is fixed for the current sampling interval. Using data available through t_k , including the measured output y_k , the controller computes u_{k+1} for application at t_{k+1} . The initial input u_0 is prescribed. The completed transition ending at t_k was generated by u_{k-1} .

3. Structure-Informed Nonlinear Models

We specify which functional relations are retained, which numerical quantities are unknown, and which coefficient relations are imposed. These specifications determine the regressor and remain fixed during online coefficient identification. The amount of supplied information and the correctness of its restrictions are distinct properties.

Table 1 summarizes the model-construction choices.

All three choices define a nonlinear predictor $\hat{F}_k(\xi, u)$. When this map is differentiable on the prediction region, we use its Jacobian-frozen affine approximation in the same constrained PCAC problem.

Table 1: Model construction from the supplied structural information.

Available information	Identification choice	Nonlinear predictor
Exact nonlinear structure with a computable parameter-linear regression.	The supplied direct, inverse, or integrated relation is retained. RLS estimates only its independent unknown parameters.	The estimates are inserted into the specified nonlinear forward model.
Trusted structural restrictions, but no complete functional model.	The supplied structure is used to construct a reduced model, such as a structure-informed linearization, Taylor/Maclaurin expansion, or other structure-informed reduced representation. Known zero terms, couplings, and coefficient relations can be used to reduce the identified parameterization.	The retained structure is combined with the fitted approximation.
Little reliable functional structure beyond a selected model class.	The predictive variables and memory are specified. A broader candidate representation, such as polynomial, radial basis function (RBF), Fourier, or other fixed nonlinear features, is selected. RLS identifies its coefficients.	The fitted nonlinear model is used.

3.1. Known functions and unknown coefficients

Consider the scalar, noise-free relation

$$y_{k+1} = d + ay_k + b \sin u_k, \quad (4)$$

where d , a , and b are unknown. Suppose the functional form in (4) is supplied. Define

$$\phi(y, u) = [1 \quad y \quad \sin u]^\top, \quad \theta = [d \quad a \quad b]^\top. \quad (5)$$

Then $y_{k+1} = \phi(y_k, u_k)^\top \theta$. The nonlinear function is retained, and only its unknown coefficients are identified. The origin need not be an equilibrium because d is unknown.

More generally, define the direct output model

$$g(s, u; \theta) = g_K(s, u) + \Psi(s, u)\theta, \quad (6)$$

where g_K is a supplied known contribution, $\Psi(s, u) \in \mathbb{R}^{p \times n_\theta}$ is the supplied regressor matrix, and $\theta \in \mathbb{R}^{n_\theta}$ contains the independent unknown coefficients. Set $g_K = 0$ when no contribution is retained exactly. The retained contribution belongs to the sampled output relation; a known continuous-time term is not a known sampled contribution without a corresponding derivation. The regressor may depend nonlinearly on s and u , but it is independent of θ .

Known coefficient relations can be retained within the regressor. For example, for

$$g(y, u; \theta) = ay + b(u + cyu), \quad \theta = \begin{bmatrix} a & b \end{bmatrix}^\top, \quad (7)$$

a known c gives $\Psi(y, u) = [y \ u + cyu]$. Estimating separate coefficients of y , u , and yu relaxes the relation between the last two coefficients. Both models contain the interaction yu , but only the first retains the specified relation.

The completed measured transition supplies

$$b_k = y_k - g_K(s_{k-1}, u_{k-1}), \quad \Phi_k = \Psi(s_{k-1}, u_{k-1}). \quad (8)$$

When a constant parameter vector represents the plant exactly and the data are ideal, $b_k = \Phi_k \theta$. Otherwise, this equality has a regression residual. Selecting a finite history and a regressor does not by itself establish exact representability.

3.2. Partial structure and local approximation

Suppose less information is supplied about (4). An affine dependence on y and an odd input contribution are retained, but the sinusoidal function is not supplied. A local polynomial model about $u = 0$ is

$$g_D(y, u) = d + ay + \sum_{\substack{1 \leq j \leq D \\ j \text{ odd}}} b_j u^j. \quad (9)$$

The retained powers follow from the supplied oddness, and their coefficients are unknown. No relation between different b_j is imposed. A finite polynomial does not represent $\sin u$ exactly on a nontrivial interval. Exact representation is not required to define the approximate predictor.

For the general Taylor construction, let g be the nominal finite-history map being approximated and write $g = g_K + h$. In a direct state-based construction, x replaces s . Define

$$z = \text{col}(s, u), \quad \zeta = z - z_o, \quad n_z = \dim z, \quad (10)$$

where z_o is a fixed expansion point. Assume h is of class C^{D+1} on a neighborhood of z_o . For a multi-index $\alpha \in \mathbb{N}_0^{n_z}$, define

$$|\alpha| = \sum_{j=1}^{n_z} \alpha_j, \quad \alpha! = \prod_{j=1}^{n_z} \alpha_j!, \quad \zeta^\alpha = \prod_{j=1}^{n_z} \zeta_j^{\alpha_j}, \quad \mathcal{A}_D = \{\alpha : |\alpha| \leq D\}. \quad (11)$$

The local expansion is

$$h(z) = \sum_{\alpha \in \mathcal{A}_D} h_\alpha \zeta^\alpha + r_D(z), \quad h_\alpha = \frac{1}{\alpha!} \partial^\alpha h(z_o). \quad (12)$$

This expansion applies to the specified smooth map. It does not establish that an arbitrary partially measured plant admits such a finite-history map.

Let $\mathcal{A} \subseteq \mathcal{A}_D$ be the index set retained under the supplied structural restrictions. Define

$$\phi_{\mathcal{A}}(z) = [\zeta^{\alpha_1} \quad \cdots \quad \zeta^{\alpha_q}]^\top, \quad q = |\mathcal{A}|. \quad (13)$$

Each monomial occurs once in a fixed ordering. Let $\Theta_{\mathcal{A}} \in \mathbb{R}^{p \times q}$ collect the true retained Taylor coefficients. Then

$$g(z) = g_K(z) + \Theta_{\mathcal{A}} \phi_{\mathcal{A}}(z) + r_{\mathcal{A}}(z), \quad (14)$$

$$r_{\mathcal{A}}(z) = r_D(z) + \sum_{\alpha \in \mathcal{A}_D \setminus \mathcal{A}} h_\alpha \zeta^\alpha. \quad (15)$$

An exact structural restriction may make an omitted coefficient zero. A deliberate omission need not do so. The second term in (15) may therefore contain powers of degree no greater than D . Truncation and additional structural reduction are separate approximations [15].

Let $S_\theta \in \mathbb{R}^{pq \times n_\theta}$ encode the imposed linear coefficient relations and define

$$\text{vec}(\Theta(\theta)) = S_\theta \theta, \quad \Psi(z) = (\phi_{\mathcal{A}}(z)^\top \otimes I_p) S_\theta. \quad (16)$$

The operator vec stacks columns. The columns of S_θ are linearly independent. Fixed nonzero coefficients are included in g_K . Independent coefficients correspond to $S_\theta = I_{pq}$. Equation (6) then gives the identification model. An incorrect coefficient restriction introduces additional model discrepancy.

The expansion point is not required to be an equilibrium. The constant monomial is retained unless its value is supplied and accounted for in g_K . The fitted coefficients need not equal the derivatives in (12); they also depend on truncation, restrictions, and the identification data. A generic polynomial model relaxes the specified restrictions within a declared monomial set. Here, generic does not mean random.

3.3. Forward prediction from measured histories

The output model predicts one new output. The remaining history is shifted without identification. For $\ell \geq 2$, define

$$F(s_k, u_k; \theta) = \text{col}(g(s_k, u_k; \theta), y_k, \dots, y_{k-\ell+2}, u_k, \dots, u_{k-\ell+2}). \quad (17)$$

For $\ell = 1$, set $F(s, u; \theta) = g(s, u; \theta)$. Equation (17) defines the same shift operation for measured histories and predicted histories. Thus, the nonlinear predictor state recursion is

$$\xi_{k+1} = F(\xi_k, u_k; \theta), \quad y_{k+1} = C\xi_{k+1}. \quad (18)$$

Only the new-output block contains identified coefficients. The shift rows are known exactly.

3.4. Forward prediction from a retained physical model

When the physical state is measured, a known differential equation may instead define F through propagation over one sampling interval. The equation used for identification need not be this forward equation.

Consider the scalar nonlinear model

$$J\dot{x} = u - dx^3, \quad J > 0, \quad d \geq 0, \quad (19)$$

where J and d are unknown and the state is measured as $y_k = x_k + v_k$. Suppose the parameters are constant on $[t_{k-1}, t_k]$ and the applied input is u_{k-1} . Integration gives

$$T_s u_{k-1} = \begin{bmatrix} x_k - x_{k-1} & \int_{t_{k-1}}^{t_k} x(t)^3 dt \end{bmatrix} \begin{bmatrix} J \\ d \end{bmatrix}. \quad (20)$$

Define $\theta = [J \ d]^\top$. The trapezoidal rule approximates the interval integral from endpoint measurements. The implemented regression quantities are

$$b_k = T_s u_{k-1}, \quad \Phi_k = \begin{bmatrix} y_k - y_{k-1} & \frac{T_s}{2}(y_{k-1}^3 + y_k^3) \end{bmatrix}. \quad (21)$$

This regression is linear in θ . Measurement and quadrature errors contribute to its residual.

For a specified θ , define X by

$$\dot{X}(\tau) = \frac{u - dX(\tau)^3}{J}, \quad X(0) = x, \quad F(x, u; \theta) = X(T_s), \quad (22)$$

where u is constant for $0 \leq \tau \leq T_s$. This forward map need not be linear in θ . Its numerical evaluation may require integration. An exact continuous-time parameterization does not make an approximate quadrature or numerical flow exact.

Related physical-identification methods use relations that are nonlinear in measured motion but linear in inertia entries [13]. Each such construction must supply its measurable regression and its forward map. No usable parameter-linear regression is presumed for every nonlinear system. State-based prediction uses the measured state in specified coordinates. For a geometric state, valid local coordinates and the linear selection of tracked coordinates must be specified before applying the formulation below.

4. Online Identification and Predictor Reconstruction

4.1. Recursive coefficient update

Both constructions supply a response $b_k \in \mathbb{R}^{n_b}$ and a regressor $\Phi_k \in \mathbb{R}^{n_b \times n_\theta}$ using data available through t_k . Within a fixed parameter regime, write

$$b_k = \Phi_k \theta + \nu_k, \quad (23)$$

where ν_k contains the applicable measurement, approximation, and model-discrepancy terms. For the direct model, $n_b = p$. For (21), $n_b = 1$.

We apply the same fixed, specified row scaling to b_k and Φ_k . The identification equations below use these scaled quantities. Let $\hat{\theta}_0^{\text{rls}} \in \mathbb{R}^{n_\theta}$ and $P_0 = P_0^\top \succ 0$ be the initial values. At step $k \geq 1$, define

$$\varepsilon_k = b_k - \Phi_k \hat{\theta}_{k-1}^{\text{rls}}. \quad (24)$$

For a forgetting factor $\lambda_k \in (0, 1]$, the update is

$$L_k = \lambda_k^{-1} P_{k-1}, \quad (25)$$

$$P_k = L_k - L_k \Phi_k^\top (I_{n_b} + \Phi_k L_k \Phi_k^\top)^{-1} \Phi_k L_k, \quad (26)$$

$$\hat{\theta}_k^{\text{rls}} = \hat{\theta}_{k-1}^{\text{rls}} + P_k \Phi_k^\top \varepsilon_k. \quad (27)$$

These updates are standard RLS operations [26, 25]. Shared unknown coefficients are updated jointly. Separate row-wise updates are used only when no unknown coefficient is shared across rows.

The choice $\lambda_k = 1$ disables forgetting. A constant $\lambda_k = \lambda < 1$ gives fixed forgetting. The residual-triggered rule below adapts the windowed construction in [25] to the specified regression coordinates. For residual-triggered variable-rate forgetting, choose $N_f \geq 1$, $\sigma_\varepsilon > 0$, $\eta > 0$, and $\gamma > 0$, and define

$$n_{f,k} = \min\{k, N_f\}, \quad E_k = \left(\frac{1}{n_{f,k} \sigma_\varepsilon^2} \sum_{j=k-n_{f,k}+1}^k \|\varepsilon_j\|_2^2 \right)^{1/2}, \quad (28)$$

$$\lambda_k = \begin{cases} (1 + \eta \min\{E_k, \gamma\})^{-1}, & E_k > 1, \\ 1, & E_k \leq 1. \end{cases} \quad (29)$$

The residual scale σ_ε is specified in these fixed regression coordinates. Forgetting changes the influence of past data; it does not change the regressor or generate excitation.

4.2. Reconstructed nonlinear predictor

Let $\mathcal{T} \subseteq \mathbb{R}^{n_\theta}$ be the specified admissible parameter set, and let $\Pi_{\mathcal{T}}$ be a specified map into that set. Define

$$\hat{\theta}_k = \Pi_{\mathcal{T}}(\hat{\theta}_k^{\text{rls}}), \quad \hat{F}_k(\xi, u) = F(\xi, u; \hat{\theta}_k). \quad (30)$$

We use the identity map when no admissibility restriction is required. For (22), a specified set $\mathcal{T} = \{(J, d) : J \geq J_{\min}, d \geq 0\}$ with $J_{\min} > 0$ retains the prescribed parameter signs and excludes a zero denominator. The unconstrained estimate remains the RLS state, whereas the mapped estimate is used for prediction.

The nonlinear one-step output prediction is

$$\hat{y}_{k+1|k} = C\hat{F}_k(\xi_k, u_k). \quad (31)$$

The estimate is updated before this prediction is constructed. The map $\hat{F}_k$ contains both the retained known structure and the current estimates of its unknown coefficients.

Parameter linearity does not establish parameter convergence. Measured variables may enter both b_k and Φ_k , so ν_k need not be independent of Φ_k [27]. Changing dynamics and incomplete model classes turn identification into a tracking or approximation problem. These effects do not alter the definition of the reconstructed predictor.

5. Local Affine Prediction and Constrained PCAC

5.1. Local affine predictor

Assume that $\hat{F}_k$ is differentiable on the region used for prediction. At the measured point (ξ_k, u_k) , define

$$A_k = \left. \frac{\partial \hat{F}_k}{\partial \xi} \right|_{(\xi_k, u_k)}, \quad B_k = \left. \frac{\partial \hat{F}_k}{\partial u} \right|_{(\xi_k, u_k)}, \quad (32)$$

where $A_k \in \mathbb{R}^{n \times n}$ and $B_k \in \mathbb{R}^{n \times m}$. The estimate $\hat{\theta}_k$ remains fixed during differentiation. Define

$$c_k = \hat{F}_k(\xi_k, u_k) - A_k \xi_k - B_k u_k. \quad (33)$$

The local affine map is

$$\hat{F}_k^a(\xi, u) = c_k + A_k \xi + B_k u. \quad (34)$$

This affine map matches $\hat{F}_k$ and its Jacobian at (ξ_k, u_k) . No equilibrium condition is used. In a Taylor-informed model, z_o remains the fixed polynomial expansion point; (ξ_k, u_k) is the current Jacobian evaluation point.

For (4), with $\xi_k = y_k$, the coefficients are

$$A_k = \hat{a}_k, \quad B_k = \hat{b}_k \cos u_k, \quad c_k = \hat{d}_k + \hat{b}_k(\sin u_k - u_k \cos u_k). \quad (35)$$

Thus, the known nonlinear function enters both the input derivative and the affine offset. For a history-based predictor, the linear shift rows in (17) are retained exactly by this construction.

5.2. Frozen finite-horizon prediction

We choose $N \geq 2$ and initialize the prediction horizon with

$$\xi_{0|k} = \xi_k, \quad u_{0|k} = u_k. \quad (36)$$

For $i = 0, \dots, N - 1$, the prediction constraints are

$$\xi_{i+1|k} = c_k + A_k \xi_{i|k} + B_k u_{i|k}, \quad y_{i+1|k} = C \xi_{i+1|k}. \quad (37)$$

The matrices A_k , B_k , and the vector c_k remain fixed over this horizon and are recomputed at the next sampling instant.

Local linearization defines the affine approximation. Freezing it over the horizon is a separate computational choice that preserves affine prediction constraints. The resulting predictions need not match recursive propagation of $\hat{F}_k$ away from the current point. If the complete reconstructed map is affine, the approximation reproduces that map. A degree-one unknown part alone does not give an affine model when a retained known part is nonlinear.

Local-linearization GPC already uses an identified nonlinear model to construct a prediction model at the current operating point. Nørgaard et al. [28] derive local derivatives and a linearization offset from a neural NARX model for GPC. González Cisneros and Werner [29] formulate a quasi-LPV NMPC procedure that solves a sequence of QPs at each sampling instant. Here, one current-point affine model is retained over each prediction horizon, without inner updates of the horizon model. Section 6.5 evaluates the resulting affine-freezing error separately from nonlinear-model error.

5.3. Constrained quadratic program

Let $r_{i|k} \in \mathbb{R}^p$ be the supplied reference over the horizon. Define

$$\Delta u_{i|k} = u_{i|k} - u_{i-1|k}, \quad i = 1, \dots, N - 1. \quad (38)$$

We specify input bounds $H_u u \leq h_u$ and input-increment bounds $H_\Delta \Delta u \leq h_\Delta$, and choose u_0 within the input bounds. Optional output bounds $H_y y \leq h_y$ have n_c rows and are softened with $\epsilon_{i|k} \in \mathbb{R}_{\geq 0}^{n_c}$.

Let $Q_i \in \mathbb{R}^{p \times p}$ satisfy $Q_i \succeq 0$, $R \in \mathbb{R}^{m \times m}$ satisfy $R \succ 0$, and $S_\epsilon \in \mathbb{R}^{n_c \times n_c}$ satisfy $S_\epsilon \succ 0$. Define $\|v\|_M^2 = v^\top M v$. The controller solves

$$\begin{aligned}
& \underset{\substack{\xi_{1|k}, \dots, \xi_{N|k}, \\ u_{1|k}, \dots, u_{N-1|k}, \\ \epsilon_{1|k}, \dots, \epsilon_{N|k}}}{\text{minimize}} & \frac{1}{2} \sum_{i=1}^N \|C \xi_{i|k} - r_{i|k}\|_{Q_i}^2 + \frac{1}{2} \sum_{i=1}^{N-1} \|\Delta u_{i|k}\|_R^2 + \frac{1}{2} \sum_{i=1}^N \|\epsilon_{i|k}\|_{S_\epsilon}^2 \\
& \text{subject to} & (36), (37), \\
& & H_u u_{i|k} \leq h_u, \quad i = 1, \dots, N-1, \\
& & H_\Delta \Delta u_{i|k} \leq h_\Delta, \quad i = 1, \dots, N-1, \\
& & H_y C \xi_{i|k} \leq h_y + \epsilon_{i|k}, \quad i = 1, \dots, N, \\
& & \epsilon_{i|k} \geq 0, \quad i = 1, \dots, N.
\end{aligned} \tag{39}$$

The terminal tracking weight is included in Q_N . When output bounds are omitted, their constraints, the slack variables, and the slack cost are also omitted. The input $u_{0|k}$ is fixed; only $u_{1|k}, \dots, u_{N-1|k}$ are optimized.

The objective is quadratic and the constraints are affine. We use a standard QP solver and apply

$$u_{k+1} = u_{1|k}^*. \tag{40}$$

This is the PCAC predictive-control step [1, 24]. The QP optimizes the frozen-affine predictor, not the nonlinear plant. Predicted output bounds do not certify actual plant output bounds.

For the bounds used in the numerical studies, a feasible QP point can be constructed whenever the committed input satisfies the hard input bounds and the prediction state and affine coefficients are finite. Set $u_{i|k} = u_k$ for $i = 1, \dots, N-1$. The resulting zero input increments satisfy the implemented increment bounds. When output constraints are included, propagate the affine predictor and set $\epsilon_{i|k} = \max\{0, H_y C \xi_{i|k} - h_y\}$ componentwise. The output slacks have no upper bound, so this choice satisfies the softened constraints. This construction does not guarantee finite model evaluations at every later step, successful numerical solution, closed-loop stability, or satisfaction of actual plant-output bounds.

At each step, the controller processes the completed measurements, updates the unknown coefficients, reconstructs $\hat{F}_k$, evaluates A_k , B_k , and c_k ,

and solves (39). Structural information determines what is retained and what is identified. Jacobian freezing determines how the reconstructed nonlinear model is used for predictive optimization.

6. Numerical Studies

6.1. Study Design and Evaluation Protocol

The numerical investigation examines how supplied structural information and model complexity affect identification, prediction, and closed-loop control. The comparisons first consider stationary plants, then parameter variation and changes within and outside the retained model class. A physical example examines the distinction between identification and forward reconstruction. The final diagnostics separate nonlinear-model error from the error introduced by affine freezing. Table 2 summarizes the comparisons.

Table 2: Organization of the numerical studies.

Study	Section	Question	Comparison
1	6.2.1	What does supplied structure contribute?	Function choice and coefficient sharing at controlled dimension
2	6.2.2	When does a richer model help?	Polynomial degree, restrictions, and operating range
3	6.3.1	How does forgetting affect adaptation?	Gain steps and drift under three forgetting rules
4	6.3.2–6.3.3	Can retained terms accommodate new nonlinear behavior?	No change, activation of a retained term, and an out-of-class change
5	6.4	How does physical identification support prediction?	Integrated identification and direct sampled-map models
6	6.5	What changes after affine freezing?	True, learned nonlinear, and frozen-affine forecasts under identical inputs

Common settings. The studies use two scalar sampled recurrences and one scalar differential equation, specified below. For each plant,

$$y_k = x_k + v_k, \quad \xi_k = y_k, \quad C = 1. \quad (41)$$

The controller uses the measured state; the true state is used only for evaluation. All models use the identification and control procedures of Sections 4 and 5, including the committed input u_k and the computation of u_{k+1} . The dictionaries remain fixed during each experiment; the regressors below specify the retained terms and coefficient relations.

The known-model reference (K) uses the current true parameters, the same measured state, and the same frozen-affine QP. It has no knowledge of future plant changes. For Plant C, it uses the same numerical predictor integrator as the integrated physical model. This reference is diagnostic, not a performance bound.

The common settings are

$$\begin{aligned} T_s &= 0.1 \text{ s}, & N &= 10, \\ Q_i &= 1 \quad (1 \leq i < N), & Q_N &= 10, \\ R &= 0.05, & S_\epsilon &= 10^4 I_2. \end{aligned} \quad (42)$$

A separate sensitivity comparison in Appendix A (Table A.18) varies the input-increment penalty, normalized initial covariance scale, and initialization-record length one at a time. The settings of the primary comparisons remain unchanged.

States and inputs are normalized by one unit of the corresponding plant variable. Table 3 gives the hard input and input-increment bounds, soft output bounds, and durations. Within each comparison, the weights, bounds, reference preview $r_{i|k} = r(t_k + iT_s)$, and initial conditions are identical. Control runs start from $x_0 = u_0 = 0$, except in the constraint audit. No additional online probing is applied.

Identification and evaluation. Except in the sensitivity comparison, each adaptive model is fitted to the same separate 200-transition record from the initial plant regime, with $\lambda_k = 1$. Its fitted coefficients and covariance initialize the control run after the plant is restarted. The prior-frozen control is the stated exception. Independent 600-transition records evaluate prediction accuracy after 25, 50, 100, and 200 fitting transitions. These tests use

Table 3: Bounds and run durations in each plant’s stated units.

Plant	$u_{\max}$	$\Delta u_{\max}$	$x_{\max}$	Duration (s)
A	1.0	0.25	1.2	120
B	1.2	0.25	1.2	80
C	1.0	0.10	1.2	80
B: constraint audit	1.2	0.25	0.85	10

common data, whereas subsequent closed-loop identification data can differ because the controllers apply different inputs.

Fixed row and column scalings are applied before RLS. In the normalized parameter coordinates, the initial covariance is $P_{\beta,M,\text{init}} = (100/n_{\theta,M})I_{n_{\theta,M}}$, where $n_{\theta,M}$ is the number of independent coefficients. This convention equalizes average prior regression variance on the specified calibration grids, not physical parameter uncertainty across different identification relations. The initial models are specified with each study. Appendix A gives the input records, scaling definitions, and calibration grids.

The primary comparisons are noise-free. For the stationary study of supplied structure, 40 paired trials use measurement noise $v_k \sim \mathcal{N}(0, 0.01^2)$ and $w_k = 0$. The selected gain-step comparison uses 40 paired trials with the same measurement noise and $w_k \sim \mathcal{N}(0, 0.005^2)$. The sequences are mutually and temporally independent and are paired across models by sample index. The physical study remains deterministic. The trial selections and independent initialization and control noise streams are specified in the appendix.

Performance measures and checks. The principal measures are tracking and nonlinear prediction root-mean-square errors (RMSEs), input RMS, and mean absolute input increment. Tracking is scored from $x_k - r_k$ over fixed evaluation windows. One-step prediction is scored against the next state, before that transition is used for identification. Parameter error is evaluated only for models with an exact reference parameter vector. Section 6.5 defines the nonlinear and affine forecast errors. The appendix gives the complete metric definitions, evaluation windows, recovery criteria, and parameter scales.

Numerical checks cover reconstruction, the RLS update, input timing, predictor derivatives, integration, and QP residuals. Predicted slack and actual bound violations are recorded separately. Rejected identification updates retain the last valid estimate and covariance; rejected control computations hold the input. Runs terminate if the state becomes nonfinite or

its magnitude exceeds five state units. All attempted trials and failures are reported; incomplete trajectories are not scored as complete runs. Repeated-trial summaries use medians, interquartile ranges, and paired differences, with uncertainty calculations specified in the appendix.

6.2. Structural Information and Model Complexity

This subsection addresses the first two questions in Table 2. The first comparison changes supplied information at fixed or explicitly varied parameter dimension. The second changes approximation order and operating range for a non-polynomial plant.

6.2.1. Supplied structure at fixed parameter dimension

Consider Plant A,

$$x_{k+1} = d_A + a_A x_k + b_{A,k}(1 + c_A x_k)u_k + h_k x_k^2 + \rho_k \sin(2x_k) + w_k. \quad (43)$$

The state and input are dimensionless, and the recurrence defines the sampled plant directly. For the stationary comparison, we set

$$(d_A, a_A, c_A) = (0.03, 0.65, 0.35), \quad b_{A,k} = b_A = 0.45, \quad h_k = \rho_k = 0. \quad (44)$$

Table 4 lists the compared adaptive models. The shared model (S) enforces the correct coupling relation with $c_A = 0.35$. The incorrectly shared model (W) enforces the opposite sign. The relaxed model (R) and the complete quadratic model (P₂) fit the relevant coefficients independently. No adaptive model receives the true values of d_A , a_A , or $b_{A,k}$.

Table 4: Direct models for stationary Plant A.

Model	Complete regressor row $\Psi_M(\xi, u)$	$n_{\theta, M}$
Affine (A)	$[1, \xi, u]$	3
Shared (S)	$[1, \xi, (1 + c_A \xi)u]$	3
Incorrectly shared (W)	$[1, \xi, (1 - c_A \xi)u]$	3
Relaxed (R)	$[1, \xi, u, \xi u]$	4
Complete quadratic (P ₂)	$[1, \xi, u, \xi^2, \xi u, u^2]$	6

The exact stationary coefficient vectors are

$$\begin{aligned} \theta_S^* &= (d_A, a_A, b_A)^\top, \\ \theta_R^* &= (d_A, a_A, b_A, b_A c_A)^\top, \\ \theta_{P_2}^* &= (d_A, a_A, b_A, 0, b_A c_A, 0)^\top. \end{aligned} \quad (45)$$

The affine and incorrectly shared models do not contain the nominal plant on an open state–input region. No exact parameter target is assigned to them. The complete map of the affine model is affine.

We use $\lambda_k = 1$ and include the known-model reference. Comparing the shared model with the affine and incorrectly shared models changes functional structure or restriction correctness at the same coefficient count. Comparing the shared and relaxed models relaxes sharing within the correct set of terms. Comparing the relaxed and complete quadratic models adds terms whose true coefficients are zero.

Before fitting, we initialize the shared, relaxed, and complete quadratic models with the common map $0.5\xi + 0.25(1 + c_A\xi)u$. The affine model is initialized with $0.5\xi + 0.25u$ and the incorrectly shared model with $0.5\xi + 0.25(1 - c_A\xi)u$. All five initial maps have the same value and Jacobian at the origin.

Let t be measured in seconds and define

$$\begin{aligned} r_A(t) &= r_j, & j &= \min\{5, \lfloor t/20 \rfloor\}, \\ (r_0, \dots, r_5) &= (0.6, -0.5, 0.75, -0.4, 0.6, 0). \end{aligned} \tag{46}$$

The equilibrium input is $u_{\text{eq}}(r) = ((1 - a_A)r - d_A)/[b_A(1 + c_Ar)]$. The prescribed reference levels lie within the steady-state input authority for both $b_A = 0.45$ and 0.30 . This algebraic check does not establish transient feasibility.

Figure 1 compares held-out prediction and whole-run tracking for stationary Plant A. The known-model reference performs no identification; its held-out prediction RMSE is 2.82×10^{-17} (numerical roundoff) and is excluded from the logarithmic axis in panel (a). Some interquartile ranges are narrower than the symbols.

The shared model improves prediction without increasing the number of estimated coefficients relative to the affine model. After 200 initialization transitions, the median one-step error on the independent noise-free 600-transition evaluation record is 5.30×10^{-4} for the shared model and 1.79×10^{-2} for the affine model across the 40 noisy fits. The corresponding values are 6.63×10^{-4} for the relaxed model and 9.50×10^{-4} for the complete quadratic model. Thus, supplying the correct coefficient relation reduces the median prediction error under the specified data and initialization conditions. The incorrectly shared model has a larger error than the affine model.

The prediction differences are not proportional to the differences in closed-loop tracking. The whole-run scores cover $0 \leq t < 120$ s and retain all

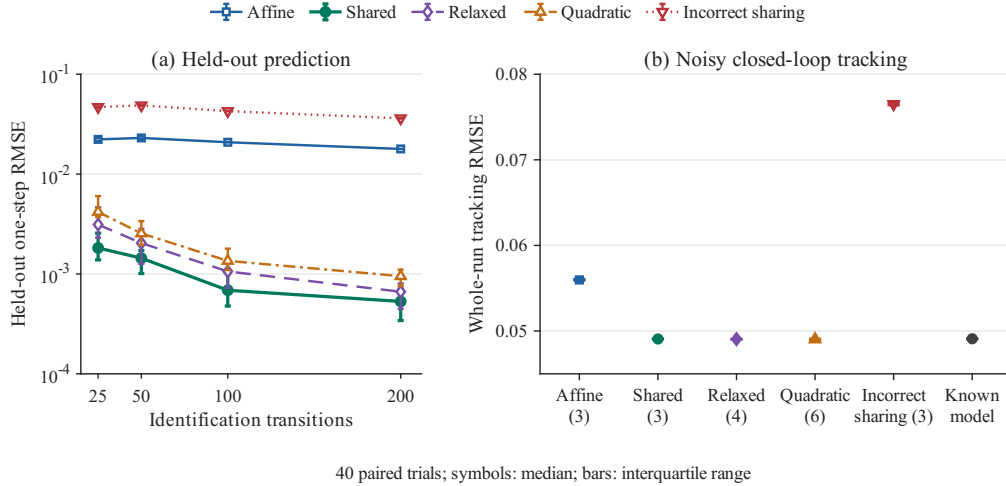


Figure 1: Study 1: structural information, identification, and closed-loop tracking. (a) Held-out one-step prediction RMSE after 25, 50, 100, and 200 identification transitions. (b) Whole-run tracking RMSE for 40 paired noisy closed-loop trials. Symbols denote medians and bars interquartile ranges. Parentheses give the number of estimated coefficients. The known-model prediction error in (a) is at numerical roundoff.

reference transitions. The median whole-run tracking RMSE decreases from 0.05597 for the affine model to 0.04906 for the shared model, a reduction of approximately 12.3%. The values for the relaxed model, the complete quadratic model, and the known-model reference are 0.04905, 0.04903, and 0.04909, respectively. The shared, relaxed, and complete quadratic models have close whole-run tracking medians in absolute terms. Additional coefficients reduce this tracking median slightly but increase the held-out prediction error in this comparison. Table A.17 reports the paired tracking differences and their intervals.

Table 5 summarizes the reported medians.

6.2.2. Approximation order and operating region

Consider Plant B,

$$x_{k+1} = 0.03 + 0.65x_k + 0.40 \sin u_k + w_k. \quad (47)$$

Table 5: Medians for stationary Plant A. One-step errors use the independent noise-free evaluation record after 200 noisy initialization transitions; tracking RMSE uses the whole control run. A dash denotes a value not tabulated here.

Model	One-step RMSE	Tracking RMSE
Affine (A)	1.79×10^{-2}	0.05597
Shared (S)	5.30×10^{-4}	0.04906
Relaxed (R)	6.63×10^{-4}	0.04905
Complete quadratic (P_2)	9.50×10^{-4}	0.04903
Known-model reference (K)	—	0.04909

This dimensionless recurrence is non-polynomial in the input. We compare the known-model reference with the direct models

$$\begin{aligned}
\Psi_A &= [1, \xi, u], & n_{\theta, A} &= 3, \\
\Psi_E &= [1, \xi, \sin u], & n_{\theta, E} &= 3, \\
\Psi_{O_3} &= [1, \xi, u, u^3], & n_{\theta, O_3} &= 4, \\
\Psi_{O_5} &= [1, \xi, u, u^3, u^5], & n_{\theta, O_5} &= 5, \\
\Psi_{P_3} &= [1, \xi, u, \xi^2, \xi u, u^2, \xi^3, \xi^2 u, \xi u^2, u^3], & n_{\theta, P_3} &= 10.
\end{aligned} \tag{48}$$

The exact-sine model (E) receives the sine function but not its coefficient. Its exact coefficient vector is $(0.03, 0.65, 0.40)^\top$. The odd cubic (O_3) and odd quintic (O_5) models receive affine state dependence and odd input dependence. Their expansion point is the origin and their odd-power coefficients are independent. Sine-series coefficient ratios are not supplied. The complete cubic model (P_3) removes these restrictions within the complete cubic dictionary. None of the polynomial models contains the sine dependence exactly on a nontrivial input interval.

We use $\lambda_k = 1$ and, before fitting, initialize all polynomial models with $0.5\xi + 0.25u$ and the exact-sine model with $0.5\xi + 0.25\sin u$. For separate control runs, define

$$r_B(t) = 0.08 + A_r \sin(2\pi 0.05t), \quad A_r \in \{0.2, 0.6, 1.0\}. \tag{49}$$

The control runs use the same initialization record and restart state for every amplitude. Only the reference amplitude changes between these control runs. The same fitted models are evaluated on the three independent input-range records specified in Section 6.1. Reference amplitude and evaluation input range are separate variables.

Table 6: Realized ranges in Study 2 over $20 \leq t < 80$ s, pooled across the six model runs at each reference amplitude. State and input are dimensionless. The final unused committed input is excluded.

A_r	Minimum state	Maximum state	Minimum input	Maximum input
0.2	-0.13899	0.29442	-0.19841	0.18416
0.6	-0.53729	0.69589	-0.57811	0.56478
1	-0.92033	1.08025	-1.08144	1.06039

Comparing the odd cubic and complete cubic models changes restrictions at the same maximum degree. Comparing the odd cubic and odd quintic models increases degree under the same supplied information. The comparison with the exact-sine model separates polynomial approximation from fitting the exact function. The input derivative satisfies $0.40 \cos u \geq 0.40 \cos(1.2) > 0$ on the actuator interval. The realized state and applied-input ranges on $20 \leq t < 80$ s are reported in Table 6. Each row gives the minimum and maximum across the six model runs at that reference amplitude; these are realized ranges, not prescribed bounds or ranges attained by every individual controller.

Results. Figure 2 separates prediction after the common 200-transition fit from tracking during subsequent online adaptation. The results are from the deterministic, noise-free comparison. Each fitted predictor is evaluated without refitting on three independent 600-transition records. Their input ranges $[-L, L]$ remain within the initialization input envelope; larger-input extrapolation is not tested. The control runs instead vary the reference amplitude, with tracking summarized over the prescribed interval $20 \leq t < 80$ s.

The exact-sine model has the lowest held-out error among the adaptive models at all three evaluation ranges. At $L = 1.1$, its RMSE is 2.095×10^{-5} , compared with 1.659×10^{-2} for the affine model, although both estimate three coefficients. The odd cubic gives 2.053×10^{-4} without retaining the exact sine function. Neither adding the fifth-order term nor using the complete cubic dictionary improves this initial prediction score. The fitted polynomial errors do not increase monotonically with the evaluation input range.

The closed-loop comparison gives a different ordering among the polynomial models. The odd quintic has lower post-transient tracking error than the odd cubic at all three reference amplitudes. At $A_r = 1.0$, the error falls from 4.356×10^{-4} to 2.368×10^{-4} , a reduction of 45.6%, but an absolute difference of only 1.988×10^{-4} state units. The complete cubic, with ten

Table 7: Study 2 at $A_r = 1.0$. The coefficient count is n_θ . Input RMS and mean absolute increment use $20 \leq t < 80$ s.

Model	n_θ	Tracking RMSE		Input RMS	Mean $ \Delta u $
		$0 \leq t < 80$	$20 \leq t < 80$		
Affine	3	2.174×10^{-2}	2.098×10^{-2}	0.7092	0.02042
Exact sine	3	4.126×10^{-3}	1.361×10^{-4}	0.7124	0.02140
Odd cubic	4	4.148×10^{-3}	4.356×10^{-4}	0.7123	0.02141
Odd quintic	5	4.136×10^{-3}	2.368×10^{-4}	0.7123	0.02140
Complete cubic	10	4.148×10^{-3}	4.023×10^{-4}	0.7124	0.02142
Known-model reference	0	4.127×10^{-3}	1.459×10^{-4}	0.7124	0.02140

estimated coefficients, does not provide a consistent tracking improvement over the four-coefficient odd cubic. Table 7 also reports the whole-run scores, which differ much less among the nonlinear models. Initial prediction and later tracking use different estimates and data; their rankings need not coincide.

At the largest reference amplitude, the exact-sine model and known-model reference give closely comparable tracking in absolute terms. The known-model reference uses the same frozen-affine predictor and QP and does not provide a theoretical lower bound on tracking error. The small difference does not establish an advantage of parameter estimation itself.

All 18 amplitude runs completed without rejected identification updates or control computations. Re-evaluation of the recorded 50-transition Gram matrices found numerical rank deficiency at the declared relative threshold 10^{-10} for the complete cubic at all three amplitudes, and for the odd quintic at the smallest amplitude. Appendix A specifies the rank criterion and reporting convention. These finite-window diagnostics concern recent data, not the accumulated RLS information, and do not establish the cause of a performance difference.

Under these noise-free conditions and the chosen initialization, useful nonlinear functions and restrictions matter more than coefficient count alone. Additional degree can improve sustained tracking without improving the initial prediction score.

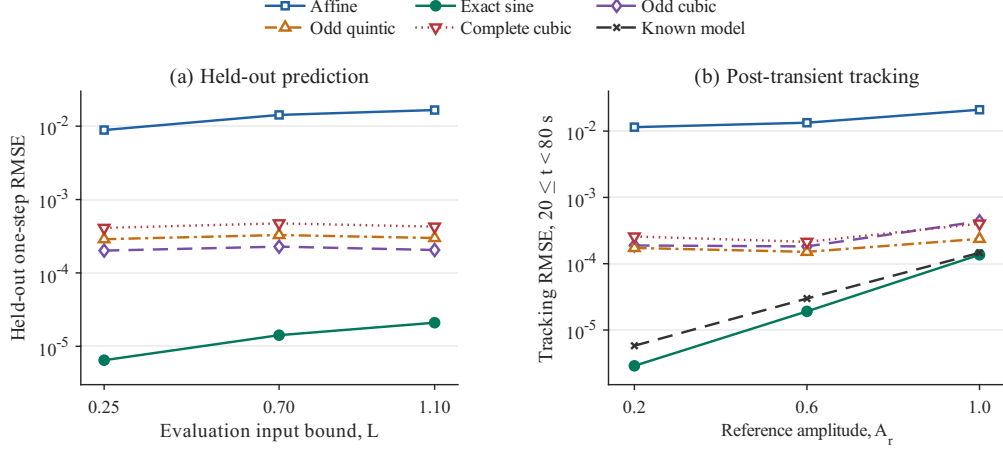


Figure 2: Study 2: approximation order and operating region. (a) Held-out one-step prediction RMSE after 200 identification transitions, versus evaluation input bound L . (b) Tracking RMSE on $20 \leq t < 80$ s, versus reference amplitude A_r . Results are deterministic and noise-free. The known-model prediction error is at numerical roundoff and is omitted from (a).

6.3. Adaptation to Plant Changes

This subsection addresses parameter variation and structural activation in Table 2. The dictionaries remain fixed. Parameter variation changes coefficients of already present plant terms. Structural activation changes the set of nonzero plant terms. The relation of each structural change to the available dictionary is specified separately.

6.3.1. Parameter variation

We use Plant A with the nominal d_A , a_A , and c_A , and set $h_k = \rho_k = 0$. For the abrupt change, define

$$b_{A,k} = 0.45 - 0.15 \mathbf{1}_{\{t_k \geq 50\}}. \quad (50)$$

In a separate experiment, define the drift

$$b_{A,k} = 0.45 - 0.15 \min\{1, \max\{0, (t_k - 50)/40\}\}. \quad (51)$$

Both changes preserve the shared representation. Both experiments use reference (46) and the same initialization fit.

Each of the affine and shared models is paired with no forgetting, fixed forgetting $\lambda_k = 0.99$, and the residual-triggered variable-rate forgetting rule in (29). The parameters of this rule are

$$N_f = 10, \quad \sigma_\varepsilon = 0.03, \quad \eta = 0.02, \quad \gamma = 10. \quad (52)$$

The residual scale is specified after the fixed row scaling. The online residual window starts empty at the control-run restart and is not cleared at the plant change.

The comparison also includes the known-model reference and two fixed shared-model controls. The *pretrained-frozen* control retains the fitted model after the common initialization record. The *prior-frozen* control retains the raw initial model without fitting or online updates. These controls separate the value of continued adaptation from the value of the initialization data.

No adaptive controller receives the event time, future parameter values, or a switch-triggered reset. A change indexed by k_c first affects x_{k_c+1} . The first changed transition reaches RLS at $k_c + 1$. The known-model reference receives the current parameter at k_c . Drift is evaluated over the changing interval $\mathcal{W}(50, 90)$ and the late interval $\mathcal{W}(90, 100)$ separately; no settling time is assigned.

Results. The study comprises nine noise-free controllers for each gain schedule and 40 paired mixed-noise realizations of the five-controller shared-model comparison for the abrupt change. All 218 runs completed without rejected identification updates or control computations. Figure 3 shows the noise-free shared-model responses. Table A.16 in Appendix A retains the complete deterministic comparison.

Following the abrupt change, both forgetting rules reduce the late-window tracking error relative to no forgetting or a model frozen after fitting. Table 8 shows that this result also holds in the mixed-noise trials. The median late-window RMSE is 0.07244 without forgetting, 0.02417 with fixed forgetting, and 0.02777 with variable-rate forgetting, compared with 0.12574 for the pretrained-frozen model. These scores cover the prescribed window and are not steady-state errors.

The event-window and recovery comparisons give different orderings. Variable-rate forgetting has a lower event-window RMSE than fixed forgetting, but this does not imply earlier recovery. In the noise-free shared-model run, fixed forgetting satisfies the prescribed tolerance and two-second dwell with a recovery delay of 5.8 s; variable-rate forgetting does not complete the

dwelling by 60 s. The event window also includes anticipation of the reference change at 60 s. In particular, the fixed-forgetting response has a larger error near that change. The declared window is retained, but its RMSE should not be interpreted as an isolated response to the gain step.

In the mixed-noise trials, none of the four fitted shared-model variants satisfies the complete recovery criterion by 60 s. These 160 outcomes are right-censored, not assigned a recovery delay of 10 s. The known-model reference recovers in all 40 trials, with delays of 0.2–0.3 s. Non-recovery under the specified dwell condition is distinct from an incomplete run or a failure of the numerical solver.

The variable-rate rule responds differently to the two change schedules. For the noise-free shared model, forgetting first activates at 50.2 s after the abrupt change, but only at 75.1 s during the drift that begins at 50 s. Over the drift interval $50 \leq t < 90$ s, tracking RMSE is 0.07349 without forgetting, 0.05221 with fixed forgetting, and 0.06579 with variable-rate forgetting. The corresponding late-window scores are 0.09833, 0.03179, and 0.04501. With the specified residual threshold, fixed forgetting gives lower tracking error for this gradual variation. No general ordering of forgetting rules follows.

Improved tracking does not imply recovery of the individual plant coefficients. Although only the input gain changes in the plant, all three shared-model coefficients remain adaptive. In the noise-free abrupt-change run, the fixed-forgetting gain estimate falls to about 0.087 even though the true gain is 0.30; the estimated state coefficient simultaneously increases to about 0.917. This compensation, together with the frequent numerical rank deficiency of recent noise-free regressor windows, is consistent with limited information for distinguishing individual coefficients. The finite-window diagnostic does not establish singularity of the accumulated RLS information or parameter convergence.

The affine and shared models also exchange rankings in the abrupt-event window. Thus, retaining the correct representation does not guarantee the smallest tracking score for every transient. The prior-frozen control has substantially larger errors and violates the soft output bound in both noise-free schedules, although its hard input bounds remain satisfied. Overall, this study supports a distinction between closed-loop readaptation, residual-triggered forgetting, and recovery of the true coefficients, even when the changing plant remains within the retained model class.

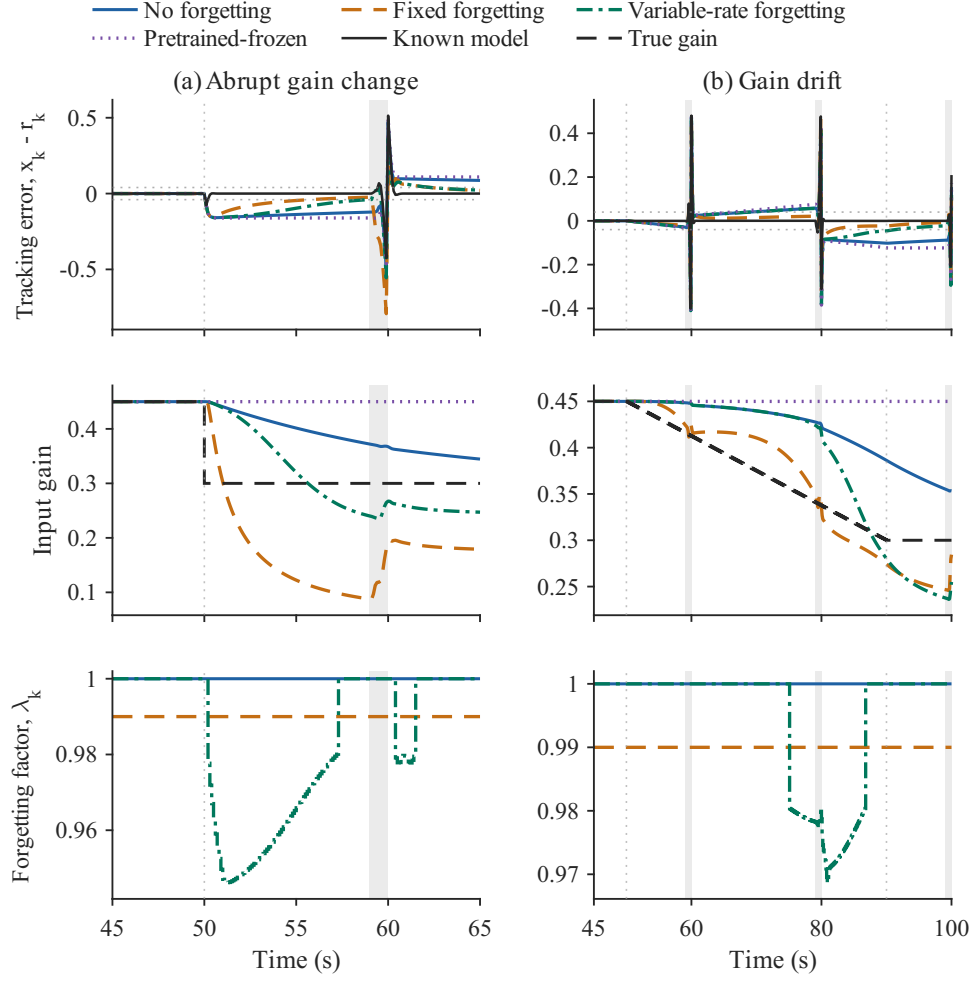


Figure 3: Study 3: noise-free shared-model responses to an abrupt gain change (left) and drift (right). Rows show tracking error, gain estimate, and forgetting factor. The known-model tracking reference and true gain are included. Shading marks reference-preview intervals.

Table 8: Study 3 mixed-noise abrupt change. Tracking entries are medians [first, third quartiles]. Recovery uses the prescribed two-second dwell. Each row contains 40 completed runs.

Controller	Tracking RMSE		Recovered by 60 s
	$50 \leq t < 60$	$90 \leq t < 100$	
No forgetting	0.14795 [0.14747, 0.14855]	0.07244 [0.07181, 0.07283]	0/40
Fixed forgetting	0.14188 [0.13818, 0.14329]	0.02417 [0.02345, 0.02464]	0/40
Variable-rate forgetting	0.12383 [0.12332, 0.12420]	0.02777 [0.02720, 0.02809]	0/40
Pretrained-frozen	0.16628 [0.16489, 0.16730]	0.12574 [0.12505, 0.12719]	0/40
Known-model reference	0.05064 [0.05006, 0.05128]	0.02299 [0.02250, 0.02362]	40/40

6.3.2. Activation of an initially unnecessary nonlinear term

For the structural-change experiments only, we specialize Plant A by setting $d_A = c_A = 0$ and $b_{A,k} = 0.45$. Thus,

$$x_{k+1} = 0.65x_k + 0.45u_k + h_k x_k^2 + \rho_k \sin(2x_k) + w_k. \quad (53)$$

The plant is linear when $h_k = \rho_k = 0$. This specialization removes the nominal $x_k u_k$ interaction; it is not used in the stationary or gain-variation studies.

The comparison includes the affine and complete quadratic models, the known-model reference, and the augmented affine model (A_+). The augmented affine model adds the single basis function ξ^2 to the affine dictionary, giving

$$\Psi_{A_+}(\xi, u) = [1, \xi, u, \xi^2], \quad n_{\theta, A_+} = 4. \quad (54)$$

Before fitting, all adaptive models are initialized with $0.5\xi + 0.25u$. The constant coefficient remains an unknown even though its true value is zero. We fit the adaptive models to a new common initialization record from the linear regime. Every adaptive model uses the variable-rate forgetting settings in (52), and all cases retain the same reference, bounds, and control weights.

Define the no-change control and the represented change by

$$\begin{aligned} \text{No change: } & h_k = 0, \quad \rho_k = 0, \\ \text{Represented change: } & h_k = h_* \mathbf{1}_{\{t_k \geq 50\}}, \quad \rho_k = 0, \quad h_* = 0.08. \end{aligned} \quad (55)$$

For each model, the same fitted initialization state is reused in both cases. The no-change control retains the same reference and reporting windows.

In the no-change and represented-change cases, the exact vectors are

$$\begin{aligned}\theta_{A_+,k}^* &= (0, 0.65, 0.45, h_k)^\top, \\ \theta_{P_2,k}^* &= (0, 0.65, 0.45, h_k, 0, 0)^\top.\end{aligned}\tag{56}$$

Thus the true ξ^2 coefficient changes from zero to h_* in the represented-change case. The other extra coefficients remain zero. The affine model is exact only in the linear regime.

Every retained coefficient is estimated from the beginning; none is released or inserted at the switch. An initially inactive term means a zero true coefficient, not a regressor that evaluates to zero or is excluded from RLS. The resulting adaptation occurs within a preallocated representation, not through online structure discovery.

The no-change control measures the consequence of carrying unnecessary terms. The represented change tests whether their availability becomes useful after the plant becomes nonlinear. The estimated ξ^2 coefficient is recorded together with common-query prediction and tracking errors.

Online model snapshots are evaluated every second on the fixed grid $x \in [-1, 1]$, $u \in [-0.7, 0.7]$, with 21 points per axis, against the current noise-free plant map. The no-change control uses the same grid. An estimated nonzero coefficient alone does not establish successful identification or a control benefit.

No-change and represented-change results. The deterministic comparison comprises four models under each of the three plant schedules. All 12 runs completed without rejected identification updates, control fallbacks, or violations of the prescribed plant-output and hard input or input-increment bounds. Table 9 summarizes the tracking, common-grid prediction, and recovery measures.

Without a plant change, all four controllers have a whole-run tracking RMSE of approximately 0.04958. After the common 200-transition initialization record, held-out prediction RMSE is 3.31×10^{-5} for the affine model, 4.41×10^{-5} for the augmented affine model, and 6.66×10^{-5} for the complete quadratic model. The additional coefficients therefore increase finite-data prediction error in this case, but produce little difference in the tracking score.

After the x^2 term is introduced, its estimates become appreciably nonzero in both enriched models (Figure 4, top left). The augmented affine and complete quadratic models satisfy the recovery criterion after 6.8 and 5.8 s, respectively; the affine model is right-censored at 60 s. Their late-window common-grid prediction RMSEs are 0.02464 and 0.02588, compared with 0.04275 for the affine model. Thus, retaining the added term is associated with earlier recovery and lower regional prediction error in this comparison.

These improvements do not imply accurate coefficient recovery or a uniform tracking advantage. At the last stored estimate, $t = 119.9$ s, the estimated ξ^2 coefficients are 0.0512 and 0.1099, compared with the true value 0.08. The complete quadratic estimate reaches 0.1904 during the initial response. The affine model also has lower late-window tracking RMSE than either enriched model. The event and late windows include anticipation of reference changes at 60 and 100 s, respectively. They are retained as specified, but their tracking scores are not pure measures of settled tracking or recovery speed.

6.3.3. Change outside the retained model class

We use the same linear-start plant and compared models, but set

$$h_k = 0, \quad \rho_k = \rho_\star \mathbf{1}_{\{t_k \geq 50\}}, \quad \rho_\star = h_\star \sqrt{\frac{1/5}{1/2 - \sin(4)/8}}. \quad (57)$$

This choice gives $h_\star x^2$ and $\rho_\star \sin(2x)$ the same root-mean-square magnitude for uniformly distributed $x \in [-1, 1]$. The scaling matches the total root-mean-square magnitude of the added terms, not their approximation errors within the retained model classes or the overall difficulty of the two tasks.

The new sine term lies outside every retained polynomial dictionary on a nontrivial state interval. Initialization, reference, forgetting, change time, and evaluation queries remain unchanged. No exact post-change parameter vector is assigned to these polynomial models. Their forward approximation and closed-loop response are evaluated rather than coefficient recovery. The three cases distinguish unnecessary capacity, activated capacity, and a change outside the retained span.

Unrepresented-change results. Following the sine addition, the augmented affine and complete quadratic models again satisfy the recovery criterion, after 7.4 and 6.3 s. The affine model remains right-censored. Nevertheless,

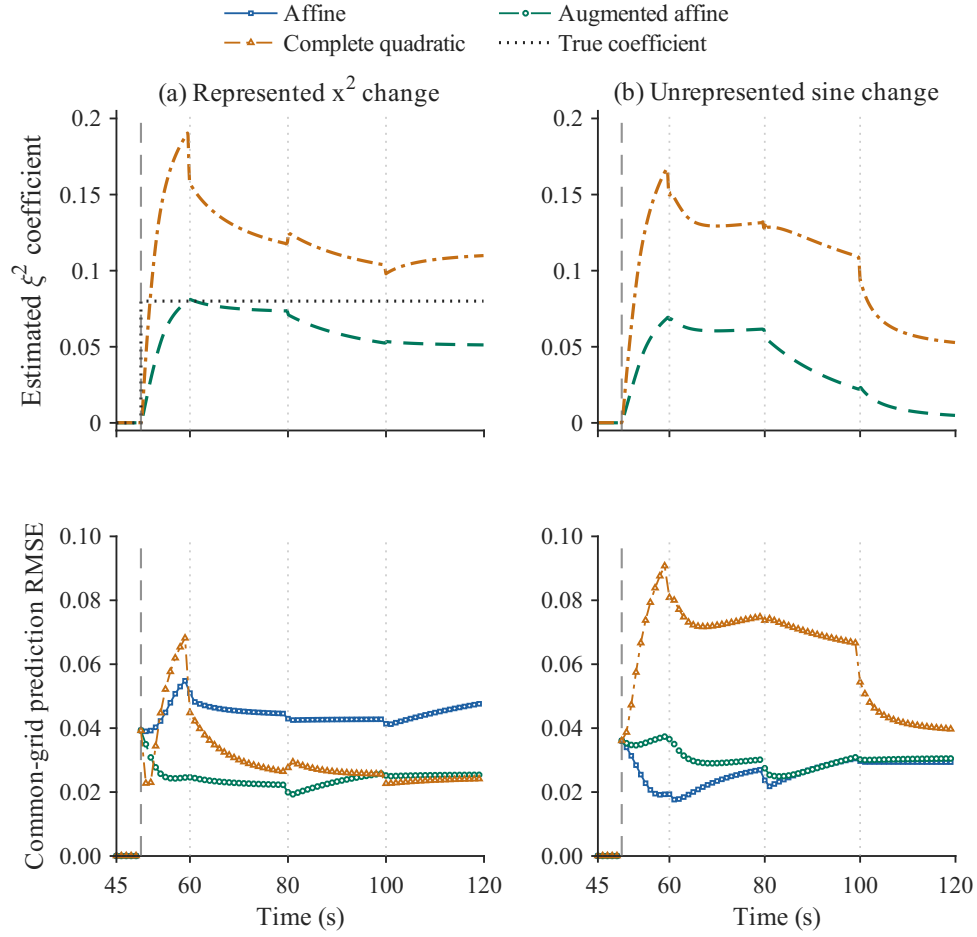


Figure 4: Study 4: represented x^2 change (left) and unrepresented sine change (right). Top: estimated ξ^2 coefficient, with its true value shown only in the represented case. Bottom: nonlinear prediction RMSE on the common grid. The plant changes at 50 s; dotted lines mark later reference changes. The known-model grid error is zero and omitted.

Table 9: Study 4 deterministic comparison. Tracking windows are the whole run $[0, 120)$, event $[50, 60)$, and late interval $[90, 100)$ s. The grid score uses the late interval. Recovery delays are measured from 50 s; censored denotes no qualifying dwell in $[50, 60)$ s, and a dash denotes not applicable.

Model	Tracking RMSE			Late grid RMSE	Recovery delay (s)
	Whole	Event	Late		
<i>No change</i>					
Affine	0.04958	0.05088	0.02111	1.040×10^{-5}	—
Augmented affine	0.04958	0.05088	0.02111	2.174×10^{-5}	—
Complete quadratic	0.04958	0.05087	0.02111	5.724×10^{-5}	—
Known-model reference	0.04958	0.05088	0.02111	0	—
<i>Represented nonlinear change</i>					
Affine	0.05398	0.08497	0.02311	4.275×10^{-2}	Censored
Augmented affine	0.05324	0.07373	0.02617	2.464×10^{-2}	6.8
Complete quadratic	0.05394	0.08774	0.03192	2.588×10^{-2}	5.8
Known-model reference	0.04923	0.04704	0.02038	0	0.2
<i>Unrepresented nonlinear change</i>					
Affine	0.05478	0.07702	0.03223	2.900×10^{-2}	Censored
Augmented affine	0.05557	0.06865	0.03511	2.906×10^{-2}	7.4
Complete quadratic	0.05581	0.07836	0.03870	6.806×10^{-2}	6.3
Known-model reference	0.04910	0.05143	0.02093	0	0.2

Each late grid score pools 441 fixed points at ten one-second snapshots. The no-change control uses the same scoring windows.

none of these polynomial models contains the changed plant exactly. The fitted ξ^2 coefficients also become nonzero in this case (Figure 4, top right), although no exact post-change polynomial coefficient vector is available. Activation of a retained coefficient is therefore not, by itself, evidence that the new plant structure has been identified.

The late-window common-grid prediction RMSE is 0.02900 for the affine model, 0.02906 for the augmented affine model, and 0.06806 for the complete quadratic model. The affine model also has the lowest whole-run and late-window tracking errors among the adaptive models in this case. Thus, the earlier recovery of the enriched models does not extend to every prediction

or tracking measure.

The grid comparison uses the same evaluation points, but the estimates are obtained from different closed-loop trajectories. Recent regressor windows are numerically rank deficient during portions of every adaptive run at the prescribed relative tolerance. This finite-window observation does not establish singularity of the accumulated RLS information. Overall, the pilot distinguishes coefficient activation, prediction over a specified region, and closed-loop recovery. It does not establish a general advantage from increasing dictionary size.

6.4. Physical Identification and Reconstruction

This subsection addresses the physical-reconstruction question in Table 2. Consider Plant C,

$$J\dot{x} = u - d_C x^3, \quad J = 1, \quad d_C = 0.6. \quad (58)$$

Let x be angular velocity in rad s^{-1} and u be torque in N m . The units of J and d_C are kg m^2 and $\text{N m}/(\text{rad s}^{-1})^3$, respectively. The parameters are constant and the input is held over each sampling interval. Only endpoint state measurements are supplied to the controller.

The exact interval relation is

$$T_s u_{k-1} = J(x_k - x_{k-1}) + d_C \int_{t_{k-1}}^{t_k} x(t)^3 dt. \quad (59)$$

For the integrated physical model (I), we use

$$b_k = T_s u_{k-1}, \quad \Phi_k = \left[y_k - y_{k-1}, \frac{T_s}{2}(y_{k-1}^3 + y_k^3) \right], \quad \theta = (J, d_C)^\top. \quad (60)$$

The identification relation has two unknown coefficients. Its endpoint implementation has quadrature error; measurement noise would also enter its regressor. Exactness of (59) does not make (60) exact.

For reconstruction, we map the raw estimate componentwise into $[0.2, 3] \times [0, 2]$, in the stated parameter units. The raw estimate and covariance remain the RLS state. Only the mapped estimates are used in

$$\dot{X} = \frac{u - \hat{d}_{C,k} X^3}{\hat{J}_k}, \quad X(0) = \xi, \quad \hat{F}_{l,k}(\xi, u) = X(T_s). \quad (61)$$

Every mapping activation is recorded. The nonlinear forward map is reconstructed from the identified physical coefficients; it is not required to be linear in them.

We compare the integrated physical, affine, and complete cubic models, the known-model reference, and the Euler-informed direct model (D)

$$\hat{F}_{D,k}(\xi, u) = \xi + \hat{\alpha}_{1,k}u + \hat{\alpha}_{2,k}\xi^3. \quad (62)$$

For the Euler-informed direct model, $g_{K,D} = \xi$ and $\Psi_D = [u, \xi^3]$, with two unknown coefficients. This reduced direct model retains an Euler-informed form but fits its coefficients to sampled data. The values T_s/J and $-T_s d_C/J$ are not exact sampled-fit targets.

The affine and complete cubic models use the regressors in (48). Only the integrated physical model has a physical parameter-error target. All models are compared through forward-output errors in common units, not through unlike inverse and direct regression residuals.

We use $\lambda_k = 1$ and $r_C(t) = 0.7 \sin(2\pi 0.05t)$. Before fitting, the integrated physical model is initialized with $(J, d_C) = (0.8, 0.3)$. The Euler-informed direct and complete cubic models are initialized with the common map $\xi + (T_s/0.8)u - (0.3T_s/0.8)\xi^3$, and the affine model with $\xi + (T_s/0.8)u$.

We use fourth-order Runge–Kutta propagation with 20 substeps for the plant and five substeps for the physical predictors of the integrated physical model and the known-model reference. Predictor derivatives are computed consistently with these stages. Both integrations require checks by refinement and by the independent zero-input solution

$$F_C(x, 0; J, d_C) = \frac{x}{\sqrt{1 + 2d_C T_s x^2/J}}. \quad (63)$$

A separate noise-free audit compares endpoint quadrature with a refined interval integral at $T_s \in \{0.1, 0.05, 0.025\}$ s on a fixed 21×21 grid over $x \in [-1, 1]$ and $u \in [-0.6, 0.6]$. The refined integral is evaluator-only information. This offline quadrature audit is not a closed-loop sampling-rate comparison.

Results. All five deterministic runs completed without rejected identification updates or control fallbacks. The physical-parameter mapping was inactive throughout initialization and control. The plant-output and hard input bounds were respected. The input-increment bound was active at three samples in each run, without a violation. Table 10 separates held-out prediction

Table 10: Study 5 deterministic comparison: prediction after 200 initialization transitions and closed-loop performance.

Model	n_θ	Held-out prediction RMSE	Tracking RMSE		Mean $ \Delta u $
			Whole	Post-transient	
Integrated physical	2	1.374×10^{-5}	2.254×10^{-3}	6.765×10^{-5}	5.031
Euler-informed direct	2	4.517×10^{-4}	2.312×10^{-3}	5.445×10^{-4}	5.039
Affine	3	6.288×10^{-3}	1.145×10^{-2}	1.098×10^{-2}	4.952
Complete cubic	10	1.468×10^{-4}	2.264×10^{-3}	1.842×10^{-4}	5.037
Known-model reference	0	2.420×10^{-11}	2.253×10^{-3}	5.788×10^{-5}	5.031

Prediction uses 600 independent evaluation transitions. Tracking windows are $[0, 80)$ and $[20, 80)$ s. All RMSEs are in rad s^{-1} . Mean input increments use $[20, 80)$ s and are expressed in 10^{-3} N m per sample.

after the common initialization record from tracking during subsequent on-line adaptation.

After 200 initialization transitions, the integrated physical model has a held-out prediction RMSE approximately 97% below that of the Euler-informed direct model, although both estimate two coefficients. The complete cubic model improves on the Euler-informed direct approximation, but uses ten coefficients and has approximately 11 times the prediction error of the integrated physical model. These comparisons concern the specified identification and reconstruction procedures, including their scaling and priors, rather than parameter count alone. The integrated physical model is not uniformly better during initial fitting: at 25 and 50 transitions, its prediction error exceeds that of the Euler-informed direct model.

After initialization, the physical estimates are $\hat{J} = 0.999995$ and $\hat{d}_C = 0.599346$ in the stated units. At the last stored estimate, $t = 79.9$ s, the values are 1.000039 and 0.599809, respectively. Both remain close to, but do not equal, the true parameters $J = 1$ and $d_C = 0.6$. Their finite-record accuracy does not establish asymptotic convergence or remove the quadrature error in the identification relation.

The endpoint-quadrature audit separates that error from numerical propagation. The RMS interval-equation residual divided by T_s is 1.180×10^{-3} , 3.284×10^{-4} , and 8.686×10^{-5} N m at sampling intervals of 0.1, 0.05, and 0.025 s, respectively (Figure 5(a)). The corresponding refined-integral RMS

residuals are below 3.2×10^{-13} N m. On the same audit grid, the largest state changes under plant and predictor substep refinement are 2.54×10^{-12} and 6.97×10^{-10} rad s⁻¹, respectively. Thus, the nonzero endpoint residual is not explained by insufficient resolution of these forward integrations. The audit varies sampling only for this numerical check, not for the closed-loop comparison.

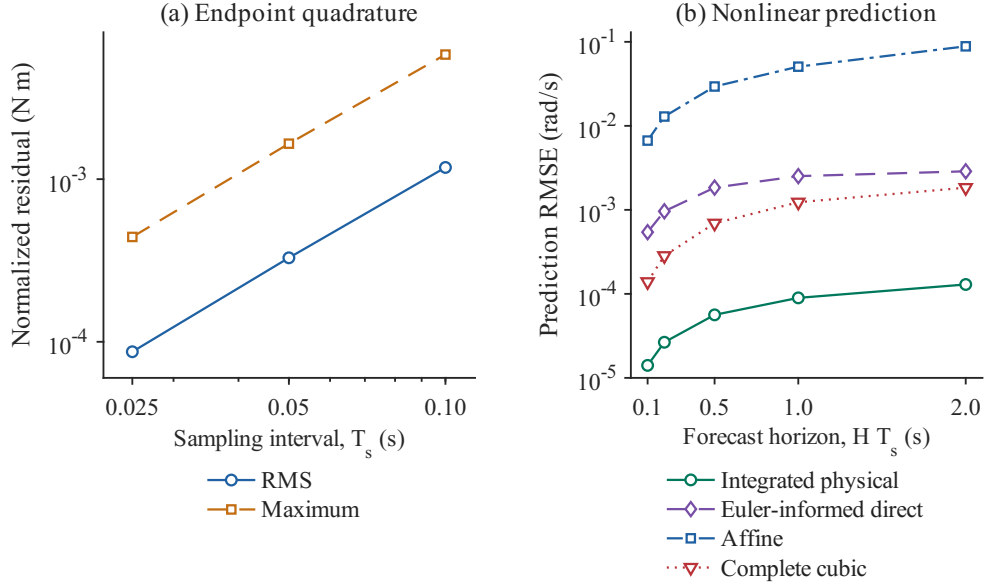


Figure 5: Study 5: (a) RMS and maximum endpoint-quadrature residual, divided by T_s , over 441 queries. (b) Nonlinear forecast RMSE after 200 initialization transitions, using 29 common initial states and rate-limited inputs. Refined-integral and known-model numerical errors are omitted from (a) and (b), respectively.

The integrated physical predictor also has the lowest nonlinear forecast error among the adaptive models at all five tested horizons, for both held and rate-limited inputs. Figure 5(b) shows the latter case using the same 29 initial states and input sequences for every model. Estimates are fixed at their 200-transition values during these forecasts. The known-model reference has a small, nonzero numerical prediction error because its integrator differs from the refined reference propagation; its rate-limited forecast RMSE remains below 1.22×10^{-10} rad s⁻¹.

Over the prescribed post-transient interval, the tracking RMSE of the

integrated physical model is about one eighth that of the Euler-informed direct model and is close to the known-model reference. The nonlinear models have similar input activity, while their whole-run tracking scores are less separated than their post-transient scores. These results support using the physical identification relation to reconstruct an accurate nonlinear predictor; they do not make the numerical implementation exact or establish a general ordering of the model classes.

The integrated physical and Euler-informed direct regressions retain full numerical rank in all 750 recorded 50-transition windows. The complete cubic regressor is numerically rank deficient in 744 of those windows at the specified relative threshold of 10^{-10} , despite completing the run. This local information diagnostic does not establish singularity of the accumulated RLS information or, by itself, explain the performance differences.

6.5. Prediction and Control Diagnostics

This subsection addresses the last question in Table 2 and reuses estimates from the preceding studies. It does not introduce another controller.

Common-query forecasts. We fix an estimator snapshot, a common initial state χ , and a common admissible input sequence $\bar{u}_0, \dots, \bar{u}_{H-1}$. Each model's affine approximation is recomputed at $(\chi, \bar{u}_0)$. Let $z_0^* = z_0^n = z_0^a = \chi$ and define

$$\begin{aligned} z_{i+1}^* &= F_k^*(z_i^*, \bar{u}_i), \\ z_{i+1}^n &= \hat{F}_{M,k}(z_i^n, \bar{u}_i), \\ z_{i+1}^a &= c_{M,k} + A_{M,k}z_i^a + B_{M,k}\bar{u}_i. \end{aligned} \tag{64}$$

Here, F_k^* is the noise-free true map with its current plant regime held fixed. For Plant C, the refined plant integrator evaluates this map. The estimates and affine coefficients remain fixed during each forecast.

We evaluate forecasts for $H \in \{1, 2, 5, 10, 20\}$ using both held-input and rate-limited input sequences from the independent evaluation records. The anchors are taken at indices 20, 40, $\dots$, 580 of each 600-transition record. Every model receives the same anchor and input sequence.

Define the terminal model, freezing, and total errors by

$$e_H^m = z_H^n - z_H^*, \quad e_H^f = z_H^a - z_H^n, \quad e_H^t = z_H^a - z_H^*. \tag{65}$$

These errors satisfy

$$e_H^t = e_H^m + e_H^f, \quad (e_H^t)^2 = (e_H^m)^2 + (e_H^f)^2 + 2e_H^m e_H^f. \quad (66)$$

RMS errors use the same query set, and the mean cross term is retained. At $H = 1$, the freezing error is zero to numerical tolerance. For a complete affine learned map, it is zero at every horizon. These identities provide checks independent of identification accuracy.

The primary diagnostic uses clean initial states and excludes future plant changes.

Forecast accuracy. The diagnostics reuse the fitted models and saved records from the preceding studies; no additional closed-loop runs are performed. Table 11 compares forecasts after 200 noise-free identification transitions at the one-second horizon, using 29 common initial states and rate-limited input sequences for each plant. The estimates and affine coefficients remain fixed during each forecast. These results are derived from existing study records, not from an independent confirmation study.

The common-query forecasts distinguish nonlinear reconstruction from the affine model used by the QP. For Plants A and B, retaining the correct nonlinear structure substantially reduces nonlinear-model error, but this advantage is reversed by affine freezing on the tested rate-limited sequences. For Plant A, the ten-step nonlinear prediction RMSE is 6.84×10^{-5} for the shared model and 4.00×10^{-2} for the affine model. After affine freezing, the error for the shared model increases to 6.53×10^{-2} , whereas the affine model is unchanged. The known-model reference has a similar frozen-prediction error.

The exact-sine Plant B model likewise has a larger total frozen-prediction error than its affine baseline. For Plant C, the integrated physical model retains a lower total error than the affine baseline, although freezing substantially reduces its nonlinear prediction advantage. Figure 6 shows the horizon dependence for the exact-sine and integrated physical models. The known-model references have nearly the same freezing errors as these accurately identified models: their total errors at one second are 0.1293 for Plant B and $0.01485 \text{ rad s}^{-1}$ for Plant C. Thus, more accurate coefficient estimates alone would not remove the observed affine approximation error.

The held-input comparisons separate this effect from identification error. For the shared Plant A model and the exact-sine Plant B model, the freezing errors remain at numerical roundoff when the input is held at its linearization

Table 11: Study 6: forecast RMSE at $HT_s = 1$ s after 200 identification transitions, under rate-limited inputs.

Plant	Predictor	Nonlinear model	Freezing	Total
A	Affine	4.003×10^{-2}	$< 10^{-15}$	4.003×10^{-2}
A	Shared	6.843×10^{-5}	6.527×10^{-2}	6.528×10^{-2}
B	Affine	3.923×10^{-2}	$< 10^{-15}$	3.923×10^{-2}
B	Exact sine	5.607×10^{-5}	1.293×10^{-1}	1.292×10^{-1}
C	Affine	5.065×10^{-2}	$< 10^{-15}$	5.065×10^{-2}
C	Integrated physical	8.963×10^{-5}	1.484×10^{-2}	1.489×10^{-2}

Each row uses the same 29 queries within its plant; evaluation input bounds are $L = 0.5$, 1.1, and 0.6 for A, B, and C, respectively. Errors are dimensionless for A and B and in rad s^{-1} for C. They are terminal forecast errors, not closed-loop tracking scores. The three columns are not additive.

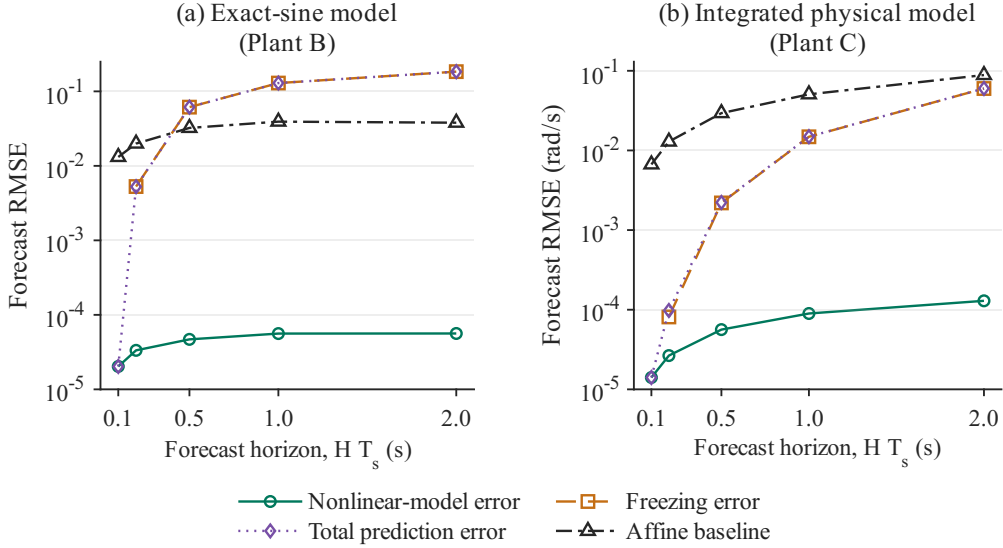


Figure 6: Study 6: nonlinear-model, freezing, and total forecast errors. (a) Exact-sine model on Plant B, $L = 1.1$. (b) Integrated physical model on Plant C. Each panel uses 200-transition estimates and 29 common initial states under rate-limited inputs; the affine baseline uses the same queries. First-step freezing errors are zero to numerical precision and omitted from the logarithmic axes.

value. At fixed input, both reconstructed maps are affine in the state, so the frozen predictor reproduces their nonlinear rollout. This property does not hold for the nonlinear state dependence of Plant C. The integrated physical model for Plant C has a held-input freezing RMSE of $1.866 \times 10^{-2} \text{ rad s}^{-1}$ at one second, compared with a nonlinear-model RMSE of $1.002 \times 10^{-4} \text{ rad s}^{-1}$.

The signed cross term in the squared-error decomposition is retained. Model and freezing errors can partly cancel; their mean-square errors cannot simply be added. The first-step freezing errors are at numerical roundoff, and complete affine models reproduce their nonlinear forecasts at every tested horizon. All archived forecast paths are finite. These diagnostics under prescribed input sequences isolate prediction effects. This fixed-model forecast comparison is separate from the receding-horizon closed-loop comparison and does not establish a closed-loop performance ordering. The receding-horizon controllers receive new measurements and reconstruct their predictors at subsequent control instants.

Initialization and unforeseen changes. Separate audits initialize learned forecasts from y_k or cross an actual plant change. Their total errors include initialization or future-change effects in addition to the terms isolated above. Neither effect is attributed to dictionary approximation or affine freezing.

The measurement-initialization audit uses only the saved Study 1 queries. Even the known-model reference has a nonzero error when prediction starts from the measured rather than the true state. For held-input forecasts, the median initialization-contribution RMSE over the 40 archived noisy trials is 6.331×10^{-3} at one step and 2.650×10^{-4} at ten steps. The nonlinear-model and freezing contributions are at roundoff in this particular known-model comparison, so the remaining error is attributable to initialization. The observed decay applies to these trajectories, not to arbitrary nonlinear plants.

The separate future-change audit uses the nine noise-free Study 3 abrupt-change cases, forecast origins 49.5 and 49.9 s, and $H = 20$, with both held and recorded applied-input sequences. Process noise is suppressed during the forecast. Only the evaluator uses the future gain schedule. For the known-model held-input forecasts, the absolute future-change contributions are 0.2514 and 0.2522 at the two origins, while model and freezing errors remain at roundoff. Knowledge of the current plant therefore does not eliminate the error due to an unannounced future change. Recorded inputs are replayed as diagnostic queries; they are not assumed known at the forecast

Table 12: Within-run terminal total forecast RMSEs for noise-free stationary Plant A. Each model uses 29 origins on its own closed-loop trajectory and the subsequently recorded applied inputs. Errors are dimensionless.

Model	$H = 1$	$H = 2$	$H = 5$	$H = 10$	$H = 20$
Affine	1.455×10^{-2}	2.552×10^{-2}	4.105×10^{-2}	4.639×10^{-2}	5.276×10^{-2}
Shared	1.700×10^{-5}	2.107×10^{-3}	1.070×10^{-2}	1.536×10^{-2}	3.590×10^{-2}
Incorrectly shared	2.756×10^{-2}	4.854×10^{-2}	8.584×10^{-2}	1.021×10^{-1}	1.439×10^{-1}
Relaxed	2.369×10^{-5}	2.102×10^{-3}	1.070×10^{-2}	1.536×10^{-2}	3.589×10^{-2}
Complete quadratic	4.928×10^{-5}	2.073×10^{-3}	1.059×10^{-2}	1.521×10^{-2}	3.573×10^{-2}
Known-model	$< 10^{-14}$	2.116×10^{-3}	1.070×10^{-2}	1.537×10^{-2}	3.590×10^{-2}

origin. These within-run queries also do not constitute a common-input comparison across controllers.

Along-trajectory forecasts. We also report stored forecasts originating on the six noise-free stationary Plant A control trajectories. For each controller, the origins are $t_k = 2, 4, \dots, 58$ s. At each origin, the measured state equals the true state, and the current online parameter estimate is held fixed during the forecast. The affine predictor is formed at that origin and is also held fixed. Nonlinear and affine forecasts use the subsequently recorded applied inputs, with terminal errors evaluated at $H \in \{1, 2, 5, 10, 20\}$. The reference is the noise-free Plant A propagation from the same origin under the same recorded inputs.

Table 12 reports RMS terminal errors over the 29 origins for each controller. At $H = 10$, the shared model has nonlinear-model, freezing, and total RMSEs of 4.433×10^{-5} , 1.537×10^{-2} , and 1.536×10^{-2} , respectively; the affine model has total RMSE 4.639×10^{-2} . These are retrospective within-run diagnostics, not the forecasts of the input plans available to the QP at each origin. The later applied inputs are replayed only for evaluation. The forecast times are common across controllers, but the states, online parameter estimates, and replayed input sequences are trajectory-specific, unlike the common-query comparisons above. The results cover the stated origins, not every sample of the 120-s runs, and do not establish a controller ordering under common state-input queries. No new closed-loop runs are used.

Constraint audit. The audit uses four dedicated Plant B constraint cases generated with Study 2, distinct from its eighteen amplitude runs. These four records are reused here without additional closed-loop simulations. These runs compare the affine, odd cubic, and exact-sine models and the known-model reference under the conditions

$$x_0 = 1.4, \quad u_0 = 0, \quad r(t) = 0.6, \quad x_{\max} = 0.85, \quad v_k = w_k = 0. \quad (67)$$

The committed input gives $x_1 = 0.03 + 0.65(1.4) = 0.94$, independently of the newly computed input. Thus the known-model QP requires at least 0.09 upper-bound slack at its first predicted output. This required slack is an algebraic consequence of timing, not a simulated result.

First-step predicted slack and actual next-sample violation are reported separately. The initial violation is recorded separately from future violations. The initial output violation is 0.55, and the first future output violates the bound by 0.09. The known-model first-step upper slack is 0.09000, whereas the affine model predicts an upper slack of 0.08544. Thus, predicted slack and actual violation are distinct even at the first step. Each of the four runs has only one future output sample outside the soft bound. These observations confirm the timing benchmark without turning the soft prediction constraints into a plant-output guarantee.

7. Discussion and Design Implications

The studies distinguish three design decisions: which functions and coefficient relations to retain, how the unknown coefficients are identified, and how the reconstructed model is used for prediction in the controller. Improving one stage does not necessarily improve the next. Table 13 summarizes the observations and their implications under the tested conditions. The resulting guidance is to use structural information to allocate model complexity, rather than to increase dictionary size indiscriminately.

Table 13: Observations and design implications of the six studies.

Study	Observation	Design implication
1	Correct sharing improved prediction at unchanged coefficient count.	Retain credible coefficient relations before adding independent unknowns.
2	Restricted polynomials were useful without exact plant representation.	Assess enrichment against the available data and intended operating region.
3	Forgetting improved readaptation without accurate recovery of individual coefficients.	Evaluate tracking, forgetting activity, and parameter error separately.
4	Nonlinear coefficients activated after represented and unrepresented changes.	Do not interpret coefficient activation as structure discovery.
5	Physical reconstruction outperformed the two-coefficient Euler-informed approximation after initialization.	Distinguish the identification relation from the forward predictor.
6	Affine freezing could reverse a nonlinear forecast advantage.	Check the horizon approximation before further enriching the dictionary.

Retain structure relevant to the task. When credible functional relations are available, they can be retained while only their independent unknown coefficients are identified. Study 1 shows that this can improve prediction without increasing the number of unknowns. When only partial information is supplied, restrictions such as oddness or known coefficient relations can instead define an approximate model. The polynomial models in Study 2 demonstrate that exact plant representation is not necessary for useful prediction and control. Sufficient local smoothness supports polynomial approximation, but it does not by itself select a useful degree or specify which coefficient relations should be imposed.

With little reliable functional information, an affine model remains a useful baseline for assessing nonlinear enrichment. In the unchanged linear case of Study 4, additional terms gave essentially no tracking benefit. The design

question is therefore whether a proposed nonlinear contribution improves the behavior relevant to the task, not simply whether it enlarges the model class. The operating region and the available identification record must be considered together. These comparisons do not determine a universal degree or a neighborhood in which an affine model suffices.

Separate representation from identification. An exact representation establishes that a suitable coefficient vector exists; it does not ensure that the available data identify that vector accurately. The retained functions must be distinguishable along the measured trajectories. Visiting a region where nonlinear terms are appreciable is not, by itself, sufficient. The compensating estimates in Study 3 and the recent-data rank diagnostics are consistent with this distinction, although they do not isolate insufficient excitation as the sole cause of the observed errors.

Forgetting reduces the influence of past data, but neither generates excitation nor introduces missing functions. Its effect also depends on the change schedule and performance measure, as the step and drift comparisons show. Similarly, the retained nonlinear terms in Study 4 aided recovery after the represented change, but also activated when approximating an unrepresented change. Tracking recovery, coefficient recovery, and identification of physical structure must therefore be assessed separately. A small tracking error does not establish accurate coefficients, and coefficient activation alone does not identify a new physical term.

Separate identification from reconstruction. Study 5 shows that retaining physical information need not require a low-order sampled-map approximation. A computable integrated relation can identify physical parameters, which are then used to propagate the nonlinear dynamics. Only the identification relation must be linear in the unknown parameters. The integrated physical and Euler-informed direct models both estimate two coefficients, but use the supplied information differently; their comparison shows why parameter count alone does not characterize a prediction model. The quadrature and propagation checks remain necessary: an exact continuous-time parameterization does not make its numerical realization exact.

Check how the controller uses the learned model. Study 6 evaluates nonlinear-model error and Jacobian-freezing error separately under common forecast queries. A single affine approximation, frozen over the horizon, can lose that accuracy as the predicted state and input depart from the linearization

point. The similar freezing errors of the known-model references show that more accurate coefficient estimates alone would not remove this error. This finding concerns the prediction approximation, not a failure of the QP solver. These forecasts under prescribed input sequences do not establish a closed-loop performance ordering, since the implemented controllers receive new measurements and reconstruct their predictors at each control instant.

On the selected noise-free Plant A trajectories, the shared model has lower total forecast RMSE than the affine model at every reported horizon in Table 12. The shared model also has a lower whole-run tracking median in the paired noisy comparison in Table 5.

When nonlinear forecasts are already accurate but frozen-affine forecasts are not, the results motivate examining the horizon approximation rather than enlarging the dictionary. Related nonlinear predictive formulations instead update state- and control-dependent models or predicted piecewise-linear model sequences iteratively over the horizon [30, 31]. A targeted future comparison would use the same identified nonlinear models with frozen-affine prediction, trajectory-dependent linearization, and nonlinear model predictive control. It would assess whether preserving more nonlinear prediction accuracy improves realized control enough to justify the additional computation. That benefit is not established here.

These implications are drawn from the reported scalar, full-state studies under the specified initialization, noise, reference, and tuning conditions. They support conditional design guidance, not a universal choice of dictionary or forgetting rule. For a new plant, the practical sequence is to retain credible structure, assess whether the data support its identification, and then check whether the controller preserves the resulting prediction accuracy. Additional complexity should address the limitation identified at that stage.

8. Conclusions

The numerical investigation examines structure-informed model construction within a common indirect adaptive predictive-control architecture. Correct functional and coefficient relations improved prediction without necessarily increasing the number of estimated parameters. Partial restrictions also supported useful approximations without exact plant representation. Physical identification provided a distinct route to accurate nonlinear reconstruction, beyond direct sampled-map fitting.

The studies also distinguish the limits of these improvements. Additional coefficients did not ensure better control; forgetting and coefficient activation did not establish recovery of the true plant parameters or structure. Accurate nonlinear prediction could, in turn, lose its advantage after affine freezing. The resulting design principle is to evaluate representation, identification, reconstruction, and horizon prediction separately, and to introduce additional complexity where the observed limitation occurs rather than treating dictionary enlargement as a general solution.

Appendix A. Numerical Protocol

Measurement and control conventions. The state, input, and output dimensions are one. The controller uses the measured state. The true state is available only to the evaluator. No observer is introduced.

Use the RLS estimator, reconstruction map, and constrained QP defined in Sections 4 and 5. At time t_k , process the transition generated by u_{k-1} and update the model. The input u_k is already committed. Initialize the prediction trajectory with $(\xi_{0|k}, u_{0|k}) = (y_k, u_k)$ and apply $u_{k+1} = u_{1|k}^*$. The model parameters and affine coefficients remain fixed within each prediction horizon.

The constraints use $|u| \leq u_{\max}$, $|\Delta u| \leq \Delta u_{\max}$, and softened $|\xi| \leq x_{\max}$.

Model conventions. Let M denote a model and let $n_{\theta, M}$ be its number of independent coefficients. A direct model has the form

$$\hat{F}_{M,k}(\xi, u) = g_{K,M}(\xi, u) + \Psi_M(\xi, u)\hat{\theta}_{M,k}. \quad (\text{A.1})$$

The regressor rows listed in Sections 6.2–6.4 are complete and ordered. Unless stated otherwise, set $g_{K,M} = 0$, use the identity admissibility map, and estimate the listed coefficients independently. Coefficient sharing is imposed within the regressor, not by estimating repeated copies of an unknown. All dictionaries are fixed before identification begins.

The known-model reference, denoted $\mathbf{K}$, uses the current true parameters, the same measurements, and the same frozen-affine QP. For Plant C, the known-model reference also uses the same numerical predictor integrator. This reference receives no future parameter schedule and is diagnostic, not a guaranteed best controller.

Initialization and common-data evaluation. For the primary comparisons, generate a separate 200-transition initialization record from each plant’s initial regime. Fit every adaptive model to this record with $\lambda_k = 1$. Carry its fitted coefficients and covariance into the control run, and restart the plant at the prescribed initial condition. Do not join the two records through a fictitious transition. No new control-run transition is available for an update at $k = 0$.

For the initialization and evaluation records, draw an input target every five samples from $L\{-1, -1/2, 0, 1/2, 1\}$ with equal probabilities. Move the applied input toward this target subject to the input-increment bound. Use initialization scales $L = 0.50, 1.10$, and 0.60 for Plants A, B, and C, respectively. Independent 600-transition evaluation records use $L = 0.50$ for A, $L \in \{0.25, 0.70, 1.10\}$ for B, and $L = 0.60$ for C.

Evaluate the estimates after 25, 50, 100, and 200 initialization transitions on the same noise-free evaluation records. These ideal-state, common-data tests separate model construction and finite-data identification from the different trajectories generated by feedback. Closed-loop comparisons do not provide identical subsequent identification data across controllers.

Scaling and initial uncertainty. Apply a fixed row scaling to each identification relation. Divide both sides of a direct regression by one output unit, and both sides of the physical inverse regression by T_s times one input unit. Let D_M contain the positive root-mean-square regressor-column magnitudes on a fixed calibration grid. Define

$$\bar{\Phi}_{M,k} = \tilde{\Phi}_{M,k} D_M^{-1}, \quad \beta_M = D_M \theta_M, \quad P_{\beta,M,\text{init}} = \frac{100}{n_{\theta,M}} I_{n_{\theta,M}}. \quad (\text{A.2})$$

Here, $\tilde{\Phi}_{M,k}$ is the row-scaled regressor. Apply RLS in the β_M coordinates and map back before reconstruction.

For direct models, use a 41×41 calibration grid over $\xi \in [-1.2, 1.2]$ and the admissible input interval. For the integrated physical model, use 41 previous-state values in $[-1.2, 1.2]$ and 41 measured increments in $[-0.1, 0.1]$, in the stated state units. These grids define scaling, not additional identification data. The convention equalizes average prior regression variance on each calibration grid, not physical parameter uncertainty across different identification relations. Raw initial models are specified in Sections 6.2–6.4.

Noise and repetitions. The primary comparisons use $v_k = w_k = 0$, where w_k is the additive process disturbance in the sampled plants. Repeat the stationary structural-information comparison with 40 paired realizations of $v_k \sim \mathcal{N}(0, 0.01^2)$ and $w_k = 0$. For the selected gain-step comparison, use 40 paired realizations with $v_k \sim \mathcal{N}(0, 0.01^2)$ and $w_k \sim \mathcal{N}(0, 0.005^2)$. Use the shared model with the three forgetting rules, its pretrained–frozen control, and the known-model reference in this second comparison.

The noise sequences are mutually and temporally independent, and are paired across models by sample index. Keep initialization inputs and references fixed across the paired trials. Initialization noise and control-run noise are independent. The physical-identification study remains deterministic.

Computational environment and random records. The saved environment records for Studies 1–5 identify MATLAB 26.1.0.3346908 (R2026a), Update 5, and Optimization Toolbox 26.1. The derived Study 6 diagnostics record the same MATLAB version and reuse the source records without generating new random streams. The initialization and evaluation input seeds are 2101 and 2102 for Study 1, 4101 and 4102 for Study 3, 5101 and 5102 for Study 4, and 6101 and 6102 for Study 5. Study 2 uses initialization seed 3101 and evaluation seeds 3102, 3103, and 3104 in increasing evaluation-range order. For noisy trial $j = 1, \dots, 40$, Study 1 uses measurement-noise seeds $2200 + j$ for initialization and $2300 + j$ during control. Study 3 uses initialization measurement/process seeds $4200 + j$ and $4300 + j$, and control measurement/process seeds $4400 + j$ and $4500 + j$. The stored bootstrap seeds are 2401 for Study 1 and 4601 for Study 3. The noise-free cases use the same input records with noise suppressed. These seeds identify the retained numerical records. The sensitivity runs used the same MATLAB and Optimization Toolbox versions, with source revision `d1f1adb0d789`. They used `quadprog` with `interior-point-convex`, constraint and optimality tolerances of 10^{-8} , and a normalized KKT rejection threshold of 10^{-7} . The paired-bootstrap calculation used `mt19937ar`, seed 7401, and 2,000 resamples. Its 95% intervals are contrast-specific, not simultaneous. These execution records identify the sensitivity calculation, not the historical source revision or solver settings of the primary studies.

Measurements. Let $K = T/T_s$, where T is the run duration, and define

$$\mathcal{W}(a, b) = \{k : 0 \leq k < K, \ a \leq kT_s < b\}. \quad (\text{A.3})$$

For a nonempty window $\mathcal{W}$, let $\mathcal{W}_\Delta = \mathcal{W} \cap \{1, \dots, K-1\}$ and define

$$\begin{aligned} E_{\mathcal{W}} &= \left(\frac{1}{|\mathcal{W}|} \sum_{k \in \mathcal{W}} (x_k - r_k)^2 \right)^{1/2}, \\ U_{\mathcal{W}} &= \left(\frac{1}{|\mathcal{W}|} \sum_{k \in \mathcal{W}} u_k^2 \right)^{1/2}, \\ V_{\mathcal{W}} &= \frac{1}{|\mathcal{W}_\Delta|} \sum_{k \in \mathcal{W}_\Delta} |u_k - u_{k-1}|. \end{aligned} \tag{A.4}$$

The last quantity is used only when $\mathcal{W}_\Delta$ is nonempty. Report tracking error in state units, input RMS in input units, and mean input increment per sample. The saved summaries also contain peak tracking error and total input variation, which are not tabulated here. Actual bound exceedances are recorded separately. Score the prospective prediction $\hat{F}_{M,k}(y_k, u_k)$ against x_{k+1} , not against the transition just used for fitting. Measured-output scores are recorded separately. Common-query prediction errors are defined in Section 6.5.

For stationary Plant A, pool the last 10 s of each 20-s reference plateau. For Plants B and C, use $\mathcal{W}(20, 80)$. For abrupt and structural changes, use $\mathcal{W}(30, 40)$ before the event, $\mathcal{W}(50, 60)$ during the event response, and $\mathcal{W}(90, 100)$ for the late response. Whole-run scores are reported separately.

The Study 1 late-plateau scores and the pre-event scores are retained in the saved summaries but are not tabulated here. The pooled late-plateau windows include reference-preview intervals and therefore do not isolate settled tracking.

For an abrupt change at 50 s, recovery is the first sampled time $t \geq 50$ s for which $|x_k - r_k| \leq 0.04$ holds at every sample in the half-open interval $[t, t+2)$. At $T_s = 0.1$ s, this requires 20 consecutive samples. The complete 2-s dwell interval must end no later than the reference change at 60 s; its right endpoint may equal 60 s. Recovery time is measured from 50 s, so zero is permitted if the condition holds throughout $[50, 52)$. Non-recovery is right-censored at 60 s, not assigned a recovery time of 10 s. Termination before that time is reported separately.

Parameter error is evaluated only when an exact reference vector is spec-

ified. Component errors are recorded, together with the scaled error

$$E_{\theta,M,k} = \frac{100}{\sqrt{n_{\theta,M}}} \left\| D_{\theta,M}^{-1} (\hat{\theta}_{M,k} - \theta_{M,k}^*) \right\|_2, \quad (\text{A.5})$$

where $D_{\theta,M}$ contains fixed component scales. Use one unit for each dimensionless coefficient and one stated physical unit for each physical coefficient. Raw and mapped physical estimates are recorded separately. The scaled error in (A.5) is not tabulated here.

Covariance and QP conditioning. The covariance diagnostic is $\kappa_2(P_k)$ in the normalized $\beta = D_M \theta$ coordinates of (A.2). It includes the prior, initialization record, and subsequent identification data, with the specified forgetting rule; it is not the recent-regressor Gram matrix. The QP diagnostic is $\kappa_2(H_k^{\text{QP}})$, where H_k^{QP} is the condensed Hessian including both input and output-slack variables. For these scalar $N = 10$ problems, this matrix has dimension 29: nine future inputs and twenty nonnegative output slacks. The slack penalty is included, so the diagnostic is not the condition number of the input block alone. Both condition numbers depend on the specified coordinates and weights.

Table A.14 summarizes the stored histories from 503 complete control runs. Study 1 includes six noise-free runs and 40 noisy trials for each of six models; the earlier 12 exploratory runs are excluded. Study 2 includes the 18 amplitude runs and four constraint runs; Studies 3–5 include all their reported control cases. The summaries retain every recorded control-step entry from the selected runs. All 590,000 QP condition entries and all 483,500 applicable covariance entries are finite. The 106,500 known-model covariance entries are not applicable, not failed measurements. Fixed-model controls in Study 3 retain their constant prior or fitted covariance and are included as such. The recorded control steps contain no rejected control computations or rejected attempted online identification updates. These observations do not establish general stability, convergence, or a causal relationship between conditioning and tracking error.

Finite-window numerical rank. For each recorded full 50-transition window, let G_k be the Gram matrix of the column-scaled identification regressors, excluding the RLS prior. Write its singular values as $s_{1,k} \geq \cdots \geq s_{n_{\theta},k} \geq 0$. With $\tau = 10^{-10}$, define the numerical rank

$$r_{\tau}(G_k) = \#\{j : s_{j,k} > \tau s_{1,k}\}. \quad (\text{A.6})$$

Table A.14: Maxima of the stored condition-number histories over the complete selected control runs. The counts n_P and n_H refer to covariance and QP entries. Covariance counts exclude known-model runs; the QP counts include them. No applicable nonfinite entry is excluded. Study 6 introduces no additional control run.

Study	Runs	n_P	n_H	$\max \kappa_2(P_k)$	$\max \kappa_2(H_k^{\text{QP}})$
1	246	246,000	295,200	8.8774×10^2	7.1685×10^4
2	22	12,300	14,800	3.3382×10^4	1.0162×10^5
3	218	211,200	261,600	8.5401×10^2	1.8568×10^5
4	12	10,800	14,400	1.3052×10^3	7.4769×10^4
5	5	3,200	4,000	1.0125×10^3	2.1738×10^5

The zero matrix has numerical rank zero. Use the reporting convention

$$\kappa_\tau(G_k) = \begin{cases} s_{1,k}/s_{n_\theta,k}, & r_\tau(G_k) = n_\theta, \\ +\infty, & r_\tau(G_k) < n_\theta. \end{cases} \quad (\text{A.7})$$

Here, $+\infty$ flags numerical rank deficiency at the stated threshold; it does not assert exact algebraic singularity. Report the rank range and the number of deficient windows, rather than interpreting extremely large finite condition estimates as precise measurements. A finite-only summary, when given, must identify the number of windows it excludes.

Re-evaluation of the saved matrices gives Table A.15. Each adaptive amplitude run contains 750 recorded full windows. The affine, exact-sine, and odd cubic models have full numerical rank in every such window at each amplitude. The known-model reference performs no identification, so its identification-rank diagnostic is not applicable. These diagnostics describe recent data. Since the experiment uses no forgetting, RLS also retains the initialization record and earlier transitions. Finite-window rank deficiency neither establishes singularity of the RLS covariance nor demonstrates a control failure. No regularization, clipping, or change to RLS is introduced by this reporting convention.

Study 3: complete deterministic comparison. Table A.16 retains every controller in the noise-free comparison. The event and late windows include the prescribed reference preview. A dash in the recovery column denotes right-censoring at 60 s, not a numerical failure. No settling time is assigned to drift.

All runs completed without rejected updates or control computations. Only the prior-frozen shared-model control exceeded the soft output bound;

Table A.15: Numerical ranks of the recorded full-window Gram matrices in Study 2. Deficient-window counts use the relative threshold 10^{-10} .

A_r	Complete cubic ($n_\theta = 10$)		Odd quintic ($n_\theta = 5$)	
	Rank range	Deficient/total	Rank range	Deficient/total
0.2	5–8	750/750	4	750/750
0.6	6–8	750/750	5	0/750
1.0	7–9	750/750	5	0/750

Table A.16: Study 3 deterministic comparison. Errors are tracking RMSE on the indicated time windows. Recovery delay is measured from 50 s.

Model and update	Abrupt change			Drift	
	[50, 60)	[90, 100)	Recovery (s)	[50, 90)	[90, 100)
Affine: no forgetting	0.12858	0.07393	—	0.07526	0.09939
Affine: fixed	0.14912	0.03587	5.6	0.05471	0.03848
Affine: variable-rate	0.11578	0.03977	—	0.06734	0.05253
Shared: no forgetting	0.14753	0.07184	—	0.07349	0.09833
Shared: fixed	0.14250	0.02605	5.8	0.05221	0.03179
Shared: variable-rate	0.12444	0.02899	—	0.06579	0.04501
Shared: pretrained–frozen	0.16619	0.12599	—	0.08071	0.12598
Shared: prior–frozen	0.56963	0.38548	—	0.50795	0.38550
Known-model reference	0.05009	0.02200	0.2	0.04788	0.02200

its maximum excess was approximately 0.5486 in both schedules. No applied hard input or input-increment violation was recorded.

The 50-transition normalized Gram matrices use the same numerical-rank threshold as Study 2. In the noise-free abrupt-change runs, the fractions of full windows classified as rank deficient are 65.8%, 56.8%, and 58.7% for the shared model with no, fixed, and variable-rate forgetting, respectively. These diagnostics concern recent data and do not replace the covariance or accumulated-information checks.

Verification and reporting boundaries. The saved verification records associated with Studies 4, 5, and 6 contain 97, 108, and 118 tests, respectively,

with all tests marked passed in both their before- and after-study lists. The saved Study 6 diagnostic report contains 27,296 finite forecast paths and no failure record. Its largest error-decomposition identity residual is 2.23×10^{-16} or smaller, its largest squared-identity residual is 5.79×10^{-16} or smaller, and its largest first-step freezing error is 1.12×10^{-16} or smaller. The physical quadrature and integration discrepancies are reported separately in Section 6.4. None of these numerical checks requires a particular performance ordering.

A rejected RLS update retains the last valid estimate and covariance. A rejected control computation applies $u_{k+1} = u_k$. Terminate a run when the true state is nonfinite or $|x_k| > 5$ in the stated state units. Record every rejection, fallback, and termination. Report finite-prefix scores as incomplete, not as complete-run averages. Soft output constraints are not plant-safety certificates.

For the repeated-trial comparisons, Table A.17 reports selected median within-pair differences and the stored 95% paired-bootstrap intervals from 2,000 resamples. A difference is the first-listed score minus the second-listed score; its median need not equal the difference of the two marginal medians. Every listed contrast has 40 attempted, complete, and finite pairs. The intervals are contrast-specific summaries, not simultaneous confidence intervals. The prediction contrasts use the independent evaluation record after 200 noisy fitting transitions. Tracking contrasts retain the stated whole-run, event, and late windows, including their reference-preview effects. No new bootstrap samples or closed-loop runs are generated for this reporting addition.

Sensitivity to tuning and initialization. We vary three settings of the stationary Plant A comparison one at a time. Let $\alpha_P > 0$ multiply the normalized initial covariance in (A.2), so that $P_{\beta,M,\text{init}} = \alpha_P(100/n_{\theta,M})I_{n_{\theta,M}}$, and let n_{init} be the number of initialization transitions. The nominal setting is $(R, \alpha_P, n_{\text{init}}) = (0.05, 1, 200)$. The six alternatives change only R to 0.025 or 0.10, only α_P to 0.1 or 10, or only n_{init} to 50 or 100. All five adaptive models in Table 4 are tested without forgetting. The known-model reference is evaluated only at the three distinct values of R , since it performs no identification.

The comparisons reuse the stationary study’s input records and 40 paired measurement-noise realizations. Shorter initialization uses the corresponding prefix of each record and transfers both the fitted coefficients and covariance

Table A.17: Selected paired RMSE differences from the existing noisy comparisons. Every row has 40 complete finite pairs and 2,000 stored bootstrap resamples. Differences are in dimensionless state units. Positive differences indicate a smaller score for the second-listed case, not a general controller advantage.

Contrast	Median	95% lower	95% upper
<i>Study 1 prediction after 200 fitting transitions</i>			
Shared – affine	-1.7300×10^{-2}	-1.7452×10^{-2}	-1.7212×10^{-2}
Shared – incorrect sharing	-3.5627×10^{-2}	-3.5784×10^{-2}	-3.5533×10^{-2}
Shared – relaxed	-6.1368×10^{-5}	-1.2584×10^{-4}	-3.1880×10^{-5}
Relaxed – complete quadratic	-2.6795×10^{-4}	-3.6532×10^{-4}	-1.4598×10^{-4}
<i>Study 1 tracking $[0, 120) s$</i>			
Shared – affine	-6.8967×10^{-3}	-6.9616×10^{-3}	-6.8545×10^{-3}
Shared – incorrect sharing	-2.7432×10^{-2}	-2.7530×10^{-2}	-2.7369×10^{-2}
Shared – relaxed	1.9365×10^{-5}	7.6787×10^{-6}	3.0619×10^{-5}
Relaxed – complete quadratic	2.1519×10^{-5}	-1.1491×10^{-5}	5.0351×10^{-5}
<i>Study 3 tracking $[50, 60) s$</i>			
No forgetting – fixed	6.6423×10^{-3}	5.2513×10^{-3}	7.7218×10^{-3}
No forgetting – variable-rate	2.4174×10^{-2}	2.3894×10^{-2}	2.4422×10^{-2}
<i>Study 3 tracking $[90, 100) s$</i>			
No forgetting – fixed	4.8108×10^{-2}	4.7894×10^{-2}	4.8358×10^{-2}
No forgetting – variable-rate	4.4617×10^{-2}	4.4315×10^{-2}	4.4824×10^{-2}

Table A.18: Stationary Plant A sensitivity: medians over 40 paired noisy trials. Only the indicated setting differs from the nominal $(R, \alpha_P, n_{\text{init}}) = (0.05, 1, 200)$. Panel (a) reports 10^3 times the held-out prediction RMSE after initialization; panel (b) reports the unscaled tracking RMSE over $[0, 120)$ s. All errors are in dimensionless state units.

Setting	Affine	Shared	Incorrectly shared	Relaxed	Complete quadratic
<i>(a) Held-out prediction RMSE multiplied by 10^3</i>					
Nominal	17.86	0.5305	36.18	0.6633	0.9505
$R = 0.025$	17.86	0.5305	36.18	0.6633	0.9505
$R = 0.10$	17.86	0.5305	36.18	0.6633	0.9505
$\alpha_P = 0.1$	17.83	0.6240	36.15	0.7450	1.377
$\alpha_P = 10$	17.86	0.5203	36.18	0.6730	0.9492
$n_{\text{init}} = 50$	23.01	1.439	48.74	2.039	2.548
$n_{\text{init}} = 100$	20.83	0.6867	42.69	1.060	1.353
<i>(b) Whole-run tracking RMSE</i>					
Nominal	0.055965	0.049065	0.076507	0.049053	0.049029
$R = 0.025$	0.055669	0.049064	0.075576	0.049050	0.049024
$R = 0.10$	0.056404	0.049101	0.077659	0.049084	0.049052
$\alpha_P = 0.1$	0.055963	0.049069	0.076530	0.049055	0.048971
$\alpha_P = 10$	0.055965	0.049064	0.076505	0.049053	0.049035
$n_{\text{init}} = 50$	0.056145	0.049061	0.076176	0.049011	0.048938
$n_{\text{init}} = 100$	0.056066	0.049065	0.076023	0.049025	0.048969

to the restarted control run. The raw initial models, scaling grids, reference, bounds, and control-run noise remain fixed. Prediction RMSE uses the same 600 clean held-out transitions after fitting; tracking RMSE uses the subsequent control interval $[0, 120)$ s. Input increments use samples $1, \dots, 1199$. The 38 model–configuration combinations give 1,558 runs: 38 noise-free and 1,520 noisy. The twelve nominal gate cases passed all 204 recorded comparisons with the saved stationary runs, with zero recorded differences. All 1,558 runs completed without rejected initialization or online updates, control fallbacks, or recorded output, input, or input-increment violations.

Table A.18 reports noisy-trial medians for the five adaptive models. At every tested setting, the shared model retains smaller prediction and tracking medians than the affine and incorrectly shared models, which estimate the same number of coefficients. Changing R leaves the post-fit prediction scores unchanged because the initialization fit is unchanged. It does affect control

activity. For the shared model, the median mean absolute input increment is 0.016964, 0.014860, and 0.012560 at $R = 0.025$, 0.05, and 0.10, respectively.

Shorter initialization increases the median held-out prediction error of every adaptive model, but its tracking effect depends on the model. For the complete quadratic model, reducing initialization from 200 to 50 transitions raises prediction RMSE from 9.505×10^{-4} to 2.548×10^{-3} , while tracking RMSE decreases from 0.049029 to 0.048938. The median paired tracking change, defined as the 50-transition score minus the nominal score, is -4.272×10^{-5} , with a 95% paired-bootstrap interval $[-1.433 \times 10^{-4}, -6.361 \times 10^{-6}]$. This median of paired differences is distinct from the difference of the two marginal medians.

Prior-covariance effects also depend on the model and score. Reducing α_P to 0.1 increases prediction error for the shared, relaxed, and complete quadratic models; the complete quadratic model nevertheless has lower tracking error. Increasing α_P to 10 does not uniformly improve both scores. These results preserve the stationary study’s distinction between prediction and tracking, but do not show that poorer prediction causes better control. Online adaptation and controller-specific data remain part of the tracking comparison. The sensitivity assessment is limited to the tested one-factor changes on reused records; it does not test interactions among settings, changing plants, or independent confirmation.

Cross-platform numerical verification. The reported numerical values are retained as the historical reference. Two complete reproductions, one on macOS and one on Windows, used the specified experiments and random records. The numerical-acceptance protocol was developed through diagnostic runs and frozen before both successful full reproductions. Agreement with the reference and independent checks of mathematical identities and solver validity are separate requirements.

Exact checks cover the specified configurations, seeds, input and noise records, dimensions, applicability masks, discrete outcomes, and source relationships. Each applicable finite numerical value v is compared with its reference v_{ref} using

$$|v - v_{\text{ref}}| \leq a_v + r_v |v_{\text{ref}}|, \quad (\text{A.8})$$

where a_v is an absolute tolerance in the quantity’s declared units and r_v is dimensionless. The tolerances in Table A.19 are applied elementwise in the original coordinates. A quantity required to be finite fails verification if it is nonfinite. Prescribed not-applicable and infinity conventions are checked

separately. The complete quantity assignments, source checks, and narrowly specified metadata exceptions are retained in the versioned verification specification. These tolerances are numerical acceptance requirements, not certified bounds on simulation error.

Table A.19: Cross-platform numerical-agreement tolerances. Absolute values are in the corresponding declared unit. The specified Study 1 rules take precedence. M_{ref} is the reference covariance or Gram matrix in the normalized coordinates.

Quantity	Absolute	Relative
Study 1 fitted coefficients and covariance, and 200-transition initialization prediction RMSE	10^{-10}	10^{-10}
Specified Study 1 state, output, input, prediction, and online-parameter histories	10^{-7}	10^{-7}
Other trajectories, forecasts, parameters, regression residuals, complete QP plans and slacks, metrics, bootstrap summaries, and resolved condition estimates	10^{-8}	10^{-7}
Calibration factors	10^{-12}	10^{-10}
Forgetting factors and residual energy, covariance and Gram entries, and physical-audit residuals and refinement differences	10^{-10}	10^{-7}
Covariance and Gram eigenvalues	$10^{-10} \max\{1, \ M_{\text{ref}}\ _2\}$	10^{-7}
Normalized QP residuals	10^{-9}	10^{-7}
Analytical-identity residuals	10^{-12}	10^{-7}

Numerical agreement cannot override a failed validity check. An accepted QP requires a positive solver exit status, finite outputs, and normalized primal, stationarity, dual, and complementarity residuals no greater than 10^{-7} . The original error-decomposition checks retain limits of $10^{-11}s_{\text{err}}$ for linear residuals and $10^{-11}s_{\text{err}}^2$ for squared residuals. Here, s_{err} is the greater of one and the largest absolute error component in the selected record, expressed in the original numerical coordinates. First-step and complete-affine freezing identities retain the absolute limit 10^{-11} . The physical state checks retain the $10^{-8} \text{ rad s}^{-1}$ limit and the original integration and refinement settings.

Directional and interval claims retain their specified sign, ordering, and zero-containment checks. Two restricted exceptions concern descriptive alge-

braic cross terms and specified Study 1 freezing-error contrasts whose exact value is zero. At one step, the latter exception requires agreement of the nonlinear and frozen-affine predictions. At longer horizons, the exception requires held inputs and both compared maps to be affine in the state. Every selected freezing residual must satisfy the existing 10^{-11} limit for both models in both records. The complete quadratic model is eligible only at one step. These exceptions do not follow from small magnitude alone and do not assert agreement of statistical significance.

Recent-regressor Gram diagnostics receive separate matrix, formation, spectral, rank, and source checks. Here, a passive diagnostic is recorded for analysis but does not enter the identification or control calculation. The supplementary rank threshold is 10^{-10} . A condition estimate is treated as resolved only when its prescribed numerical screening interval contains the stored and recomputed estimates and has normalized width no greater than 10^{-6} . The versioned verification specification defines the interval and its width normalization. Unresolved values and reductions can be retained only in the specified passive Gram cases, with the required paired rank and scientific-outcome checks satisfied. Such results are *qualified*, not numerically equal. This qualification does not waive the resolved-condition requirements for the RLS covariance or QP Hessian. These screening intervals are numerical diagnostics, not certified spectral bounds or proofs of algebraic rank.

The passive Study 2 QP-condition maximum has a separate locator rule. Each record must identify its own first maximizing index exactly. Different indices require agreement of the complete histories and maxima. For each record, the loss is the difference between its own maximum and its value at the other record's maximizing index. Both losses must lie within the sum of the corresponding elementwise allowances. No agreement of peak time is asserted.

Derived tables and figures must use the specified source records. Known 15-significant-digit CSV encoding receives a separate bounded representation allowance after source validation; rounded manuscript entries alone do not establish reproduction.

Certification records and source provenance. Both full reproductions used MATLAB 26.1.0.3346908 (R2026a), Update 5, and Optimization Toolbox 26.1. The macOS system used version 26.6.2 (build 25G83), an Apple M4 processor, Apple Accelerate BLAS (ILP64), and NAG Performance Components 1.4.1 with LAPACK 3.9.1, with a recorded maximum of

ten computation threads. The Windows system used Windows 11 Home (build 26200), an Intel Core i7-12700K processor, and Intel oneAPI Math Kernel Library 2025.3 with LAPACK 3.12.1, with a recorded maximum of twelve computation threads.

Both used `quadprog` with `interior-point-convex` and explicit constraint and optimality tolerances of 10^{-8} . Recorded effective options also include `MaxIterations` equal to 200 and `StepTolerance` equal to 10^{-12} .

The macOS full reproduction used revision `9855c14210643`, whereas the Windows full reproduction used revision `b8754a293810`.¹ Only two test files and their verification metadata differ between these revisions. The simulation and comparison routines, configurations, reference results, and effective acceptance policy are unchanged between these revisions. The separately executed amended test suite passed on each platform. The macOS full run was not repeated at the later revision.

Each full reproduction completed 2,073 control runs: 503 primary runs, 12 additional Study 1 diagnostic-support runs, and 1,558 sensitivity runs. Study 6 recomputed its diagnostics from the corresponding newly generated source runs. Both reproductions satisfied the prescribed numerical, source, table, and figure checks, with the stated Gram and claim-applicability qualifications retained. Outside the qualified Gram comparisons, the largest reported ratio of a numerical discrepancy to its allowance was 1.78×10^{-3} on macOS and zero on Windows. These ratios do not establish numerical agreement of the qualified Gram quantities. The protected reference records retained their paths, sizes, and SHA-256 hashes.

The original producing revisions and complete machine and solver records for Studies 1–6 are not established by the retained evidence. The certification revisions above identify the later reproductions, not the original generation of those historical results. The separately recorded source and solver information for the sensitivity calculation remain as specified earlier in this appendix.

¹The full source revisions are `9855c14210643cc06908fb529afd47953413ed6c` and `b8754a2938104f5843aebd60c4d51162223407f3`, respectively. The common acceptance-policy SHA-256 is `ca9101ea01a6eb093664d383a3d6bd1734598ba84aed2ee35b28cfa7412aec58`.

Code availability. The MATLAB implementation is available at <https://github.com/tamwng/tam-control-toolbox>. Version 2.0.0 is the designated release for reproduction and source traceability.

The reported studies provide exploratory numerical evidence under the stated conditions. No independent confirmation is claimed.

Acknowledgments

This research was partially supported by the MEXT SPreAD AI for Science program 2026 (Grant No. 26276994).

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